AIG-01 · AI IN PROJECT CONTROLS GUIDE

AI in project controls — the executive guide

What changes, what does not, and what a director should require before approving use

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PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

Summary

This guide sets out the Institute's position on artificial intelligence in a project controls function: what AI genuinely changes (the economics of coverage, of cycle time and of early warning), what it does not change (accountability for every number that leaves the function), and the six things a director should require before approving AI-assisted controls work. It closes with a value case netted honestly, the failure modes we see most often, and an approval gate usable in a meeting. The Institute's position throughout is that AI proposes; the professional disposes — every output must be explainable, validated and owned by a competent human.

About this document

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Contents

AI in project controls — the executive guide

In one paragraph. This guide sets out the Institute's position on artificial intelligence in a project controls function: what AI genuinely changes (the economics of coverage, of cycle time and of early warning), what it does not change (accountability for every number that leaves the function), and the six things a director should require before approving AI-assisted controls work. It closes with a value case netted honestly, the failure modes we see most often, and an approval gate usable in a meeting. The Institute's position throughout is that AI proposes; the professional disposes — every output must be explainable, validated and owned by a competent human.

Who this is for. Project controls directors, heads of PMO (project management office), commercial and finance directors sponsoring controls change, and the project controls managers who will be asked to make the decision work.

1. What you are actually approving

When a director approves AI in a controls function, they rarely think they are approving anything consequential — a licence, a pilot, a feature already embedded in a platform the organisation owns. What is being approved is narrower and more serious than that: **a change to how a number becomes trustworthy before it is signed.**

Every figure a controls function releases — a forecast, a valuation, an accrual, a reported percentage complete — carries an implicit assurance: a competent person produced it by a defensible method, from identified data, and would be able to explain it under challenge months later. AI does not remove that assurance requirement. It changes where the human effort sits: less in production, far more in verification. A function that adopts AI without moving effort into verification has not become more efficient; it has become faster at releasing unverified numbers.

That is the whole executive question, and everything below is a way of answering it in specifics.

2. What genuinely changes

Three economics move, and it is worth being precise about which three, because organisations routinely buy one and get charged for another.

Coverage becomes cheap. The binding constraint on much of controls has always been reading and checking capacity: nobody reads all sixty subcontracts, all the correspondence, every line of the ledger, every activity in a 1,500-line schedule. Machine capability that reads everything at low marginal cost changes which controls are feasible at all. This is the single largest and most underclaimed benefit — not that AI is clever, but that it does not tire. See `AIG-02 — What AI actually does in a controls function` for what coverage is worth, task by task.

Cycle time compresses — and creates review debt. Month-end assembly, variance narrative drafting, schedule health checking and document extraction all compress substantially. The compression is real. What comes with it is a new obligation: the exceptions that used to be found by the act of assembly are now found only if somebody deliberately looks. A pipeline fails silently where a human assembler would have noticed. Budget the review time, or the saving is borrowed rather than earned.

Warning arrives earlier. A model watching a trend across a portfolio can raise a signal months before a single bad reporting period forces it into a meeting. This is the benefit most worth having and the one most often wasted, because early warning is only valuable to an organisation whose governance can act on a signal that is not yet certain. If your change-control forum requires certainty before it will meet, an earlier signal will simply sit in a log.

3. What does not change

Accountability. A model cannot be accountable — it cannot be questioned, sanctioned, or asked what it was thinking. For every AI-influenced estimate, forecast, valuation or disclosure, a named person is accountable. "It was the model's output" is not a defence available to anybody in the chain, including the director who approved the tool.

The standard of care. The duty to verify a figure against source, to state assumptions, to distinguish what is measured from what is inferred, and to disclose material uncertainty is the discipline's existing standard. AI does not lower it. If anything it raises the practical bar, because a more capable model that is wrong is more convincingly wrong: fluent, internally consistent, correctly formatted and false.

The decisions that stay human. Some decisions must not be delegated to a model regardless of its measured accuracy, because the thing being decided is entitlement, appetite or accountability rather than computation. Baseline change acceptance, extension-of-time (EOT) entitlement, contingency release, revenue recognition and provisioning, and any decision about individuals all sit here. `AIG-10 — Human in the loop: what AI may and may not decide` owns the full decision-rights schedule; the method documents in this series state the specific prohibitions for forecasting, scheduling and risk.

4. Where the capability is weak

An executive who knows only the strengths will approve the wrong pilot. Current capability is weak at **causal reasoning** — it detects that cost and a variable moved together, not that one caused the other; at **genuinely novel scope**, where there is no comparable history to learn from; at **contractual judgement**, where the answer depends on entitlement rather than on language patterns; and at **multi-step arithmetic**, where plausible and correct diverge silently. It has no notion of truth, only of likelihood, which is why fabricated figures and citations — hallucinations — remain a live risk rather than a solved one. `AIG-02` maps this weakness onto specific controls tasks.

5. Six things to require before approving use

These are the Institute's recommended minimum. They are deliberately answerable in a paragraph each; if a proposal cannot answer them, it is not ready for approval rather than badly written.

5.1 The task class, not the tool. State which class of work is being automated — extraction, classification, drafting, forecasting from tabular data, anomaly detection, retrieval-grounded answering — and why that class fits this task better than a deterministic rule. Where a rule suffices, a rule is better: it is transparent, testable and cheap to audit. Approving "AI for controls" approves nothing reviewable.

5.2 A data-fitness statement. Which datasets feed it, who owns them, how they are coded, and what proportion currently fails validity, completeness and cut-off checks. Most AI failures in controls are data failures wearing a model's clothes. `AIG-03 — Data readiness` gives the tests; require the results, not the intention.

5.3 A verification design with a named checker. What is checked, by whom, at what frequency, and what proportion — per item, or by sample against a stated rule. A verification step that is described as "reviewed by the team" will not happen. Require the sampling rule and the exception threshold in writing.

5.4 A named owner and an audit trail. One person accountable for the output's performance, and a record for each material AI-assisted output of what the model produced, who reviewed it, what changed in review and why. This is the artefact that lets a number be defended a year later, in a dispute, to an auditor. The control framework that houses it is `AIG-08 — Governing AI on a project`.

5.5 A value case with the governance cost netted. Licences, integration, data remediation, training and the ongoing verification effort — all of it — set against measured savings, not vendor-claimed ones. The most common way an AI business case flatters itself is by pricing the licence and omitting the checking. §8 works one through.

5.6 A stop rule. What result would cause this to be discontinued, decided before the pilot starts. A pilot that cannot fail is not a pilot; it is a procurement with extra steps.

6. Sequencing

The order that works is policy, then pilot, then standardise, then integrate. Policy first is not bureaucratic instinct: without an approved-tool register and a data rule, a pilot generates confidentiality exposure and un-auditable outputs that must later be unpicked. Two sequencing errors are common enough to name. The first is **integrating before governing** — embedding AI in the month-end before anyone has defined who checks what, which produces speed and quiet risk in the same quarter. The second is **governing without ever piloting**: a policy written in the abstract, applied to no real workflow, that turns into paperwork nobody follows.

Between them sits an honest question a director should ask every quarter: which rung are we actually on, and what evidence supports the claim? An organisation that says "integrated" while three people use a public assistant unregistered is on the first rung, not the fourth.

7. How this goes wrong

The saving is claimed gross. A forty-per-cent faster close is reported as a forty-per-cent cost reduction. It is not: the verification, monitoring and data work the governance model requires are real recurring costs and belong in the same table. See §8.

Verification is assumed rather than resourced. The proposal says outputs will be checked; no one's objectives, capacity or job description changes; within two cycles the check has become a glance. This is the single most common failure, and it is invisible until an error survives to a board paper.

A pilot succeeds on the vendor's data. A demonstration on curated examples proves the tool works on curated examples. Evidence means the tool scored against your own data, with the correct answers established by your professionals beforehand. `AIG-11 — Evaluating AI tools` covers the due-diligence protocol.

Confidential data leaves before anyone notices. Commercially sensitive rates, contract positions, personal data and unpublished results are pasted into ungoverned tools because no register exists and no alternative was provided. The remedy is to provide a governed tool early, not to prohibit and hope.

The near-miss is never reported. A function reporting no AI incidents is more likely ungoverned than infallible: in an ad-hoc culture the hallucinated clause travels upward as fact instead of landing in an incident log. Treat the first reported near-miss as the governance working.

Deskilling arrives quietly. If AI drafts every narrative, codes every line and proposes every forecast, the verification the whole model depends on eventually falls to reviewers who have never done the work unaided. Keep some first-principles work in the function deliberately; `AIG-12 — The AI-literate controls professional` deals with how.

The wrong problem is automated. Speeding up a report that nobody uses, or a reconciliation that exists because two systems disagree, entrenches the waste. Ask what would happen if the output stopped, before asking how to produce it faster.

8. Worked example — the value case, netted

Illustrative figures. A controls function of six people, one portfolio, AI-assisted coding and reconciliation at month-end. Currency USD; effort in person-days of 8 hours; loaded cost USD 640 per person-day (8 × USD 80). All figures are for teaching, not benchmarks.

Baseline. Three staff spend six working days each per monthly cycle on assembly and reconciliation:

```
3 × 6 × 12 = 216 person-days per year
```

After. The same cycle takes three staff three and a half days each:

```
3 × 3.5 × 12 = 126 person-days per year
```

```
Gross saving = 216 − 126 = 90 person-days per year
```

The new work the model creates. Model monitoring, exception-rule maintenance, refreshing the sample of inputs used to score the model, and keeping the tool register current: 1.5 person-days per month.

```
1.5 × 12 = 18 person-days per year
```

```
Net effort saved = 90 − 18 = 72 person-days per year
```

```
Net effort saved in money = 72 × 640 = USD 46,080 per year
```

Netting the cost. Licences plus amortised integration and data remediation: USD 38,000 per year (assumed here; in a real case this is quoted and evidenced).

```
Net annual value = 46,080 − 38,000 = USD 8,080
```

Read it honestly. The cycle-time headline is genuine — `(18 − 10.5) ÷ 18 = 41.7 %` less effort per close — and the net financial value is USD 8,080, which will not survive a rounding error in the assumptions. Two things follow. First, the defensible justification is not the money; it is the 72 person-days moved from assembling numbers to interrogating them, plus whatever the earlier warning is worth, which should be argued explicitly rather than assumed. Second, the case is fragile: if governance effort is understated by half a day a month, that is `6 × 640 = USD 3,840` a year, and net value falls to `8,080 − 3,840 = USD 4,240`. A director who is shown a gross saving of USD 46,080 with no governance line is being shown the wrong number.

9. The director's approval gate

A usable list for the meeting in which the decision is taken. Six questions, one required answer each.

1. **Task class.** Which class of capability is this, and why is a deterministic rule not the better answer? (Required: the class named, and the rule alternative explicitly considered.)
2. **Data fitness.** What proportion of the feeding dataset currently fails validity, completeness and cut-off checks, and who owns the remediation? (Required: a measured figure, not an intention.)
3. **Verification.** Who checks what, how often, at what sample rate, and what happens to an exception? (Required: a named role and a written sampling rule.)
4. **Accountability.** Who owns the output's performance, and where is the trail of what the model proposed, who reviewed it and what changed? (Required: one name, one artefact.)
5. **Netted value.** What is the value with licences, integration, data work, training and verification effort all netted against measured savings? (Required: a net figure and its sensitivity.)
6. **Stop rule.** What result would cause us to discontinue this, and who decides? (Required: a threshold agreed before the pilot begins.)

A seventh question is worth asking annually rather than at approval: **where have we decided not to use AI, and why?** A function that cannot answer it has not been choosing.

Related

- `AIG-02 — What AI actually does in a controls function` — the capability map behind §2 and §4, task by task, including where the capability is genuinely weak.
- `AIG-03 — Data readiness: what AI needs before it is any use` — the tests behind the data-fitness statement required at §5.2.
- `AIG-08 — Governing AI on a project — the control framework` — the control framework that houses the register, the policy and the audit trail.
- `AIG-10 — Human in the loop: what AI may and may not decide` — the full decision-rights schedule summarised at §3.
- `AIG-11 — Evaluating AI tools — a buyer's due-diligence guide` — how to test a vendor claim against your own data before committing.

Sources and standards

- **PCI Body of Knowledge, Domain 13** — AI for project controls and project management (Institute manuscript, 2026). The source for the propose–verify–own workflow shape and the governance position summarised here.
- **PCI candidate AI-use policy** (Institute, 2026). The Institute's published position on AI in preparation, in assessment and in practice.
- **ISO/IEC 42001:2023**, Information technology — Artificial intelligence — Management system. In our own words: it describes a management-system approach to AI — policy, defined roles, risk assessment, controls over the AI lifecycle and periodic review — the same shape as any other management system. It is a useful skeleton for §5 and §6; it is not a substitute for the discipline-specific verification a controls function needs.
- **NIST AI Risk Management Framework 1.0** (National Institute of Standards and Technology, January 2023). In our own words: a voluntary framework organising AI risk work into governing, mapping, measuring and managing — helpful for structuring the evidence a director should ask for, and explicit that measurement without governance is not assurance.

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AIG-02 · AI IN PROJECT CONTROLS GUIDE

What AI actually does in a controls function

A capability map by class of work, and an honest account of where the capability is weak

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PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

Summary

A map of what artificial intelligence genuinely does inside a project controls function, organised by capability class rather than by product: extraction, classification, retrieval-grounded answering, anomaly detection, forecasting from tabular data, simulation support, drafting and code generation. Each class is matched to the controls tasks it fits, with the verification the professional still owes. The second half is the part most maps omit — where the capability is weak, why causal reasoning, novel scope and contractual judgement remain human, and how to triage a task between a rule, a model and a generative tool.

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Contents

What AI actually does in a controls function

In one paragraph. A map of what artificial intelligence genuinely does inside a project controls function, organised by capability class rather than by product: extraction, classification, retrieval-grounded answering, anomaly detection, forecasting from tabular data, simulation support, drafting and code generation. Each class is matched to the controls tasks it fits, with the verification the professional still owes. The second half is the part most maps omit — where the capability is weak, why causal reasoning, novel scope and contractual judgement remain human, and how to triage a task between a rule, a model and a generative tool.

Who this is for. Cost engineers, planners, project controls managers and PMO leads deciding where to apply AI first, and the managers who must approve their choice.

1. Map the capability, not the product

Product capability changes faster than any document can track, and a feature list is not evidence that a tool does the thing on your data. What is stable enough to plan against is the **class** of capability: what kind of problem the underlying technique is good at, what it needs as input, and how it fails. A professional chooses the class that fits the task, then finds a governed tool inside that class — and tests it. The tool test belongs to `AIG-11 — Evaluating AI tools`; the class choice belongs here, and it is where most of the value or waste is decided.

Three terms are needed and then used. **Machine learning (ML)** is the family of techniques that learn patterns from data rather than following written rules. **Generative AI** is the subset that produces new content — text, tables, code — of which large language models are the familiar instance. **Retrieval-augmented generation (RAG)** is the pattern in which relevant documents are retrieved and given to a generative model at the moment of answering, so the answer is grounded in your material and can cite it, rather than being recalled from whatever the model absorbed in training.

2. The eight classes that matter in controls

Class	What it does	Controls tasks it fits	What the professional still owes
Extraction	Pulls structured fields out of unstructured documents	Contract terms, notification windows, rates and retention from subcontracts; quantities from a bill of quantities; dates from correspondence	Open the cited clause and confirm every extracted value; reject anything ungrounded
Classification and coding	Assigns a category to a record	Coding invoice and timesheet lines to work breakdown structure (WBS) and cost element; sorting correspondence by contract event	Own the coding rules; work the exception queue; audit a sample of the confidently-coded
Retrieval-grounded answering (RAG)	Answers a question over your own document set, with citations	"What do our contracts say about notification periods?"; "what does our procedure require at gate 3?"	Open the citations; confirm the corpus is current and permissioned
Anomaly and pattern detection	Flags records that differ from the population	Duplicate invoices, out-of-tolerance postings, approval bypasses, unusual timesheet patterns	Set and justify the thresholds; investigate flags; know the miss rate as well as the hit rate
Forecasting from tabular data	Projects a value from historical and current data	Cost at completion candidates, delay likelihood, cash-flow shape, quantity growth	Choose and defend the method; confirm the assumption matches the cause; own the number
Simulation support	Runs and interprets probabilistic models	Monte Carlo on a quantified risk register; schedule risk analysis; scenario ranges	Own the input ranges, the correlations and the confidence level chosen
Drafting and summarisation	Produces first-cut language from data or documents	Variance narratives, basis-of-estimate text, meeting actions, board summaries	Recompute every figure; confirm every causal claim; delete what the data does not support
Code and query generation	Writes scripts, formulas and queries from a description	Data preparation, recurring reconciliations, report assembly, one-off analyses	Test against known cases before use; review it as code; never run it untested against live data

Two classes deserve a warning label. **Drafting** is where hallucination — confident, fluent, false content — does its damage, because prose hides a wrong figure better than a spreadsheet does. **Code generation** fails silently: generated code that is subtly wrong looks exactly like generated code that is right.

3. The map against the controls calendar

The same eight classes, arranged the way the work actually arrives.

When	The task	Class that fits	Realistic contribution
Set-up	Building the cost breakdown structure and code of accounts	None — this is design	AI can check consistency once you have decided; it cannot decide
Set-up	Normalising historical projects for estimating	Classification, extraction	Substantial: the tedium of mapping old codes to a current structure
Estimating	A parametric check estimate from analogues	Forecasting from tabular data	A challenge figure, never the estimate of record
Estimating	Drafting the basis of estimate	Drafting	First draft only; assumptions are the estimator's
Monthly	Coding and reconciling cost	Classification, anomaly detection	High value, comparatively low risk — the strongest first pilot
Monthly	Proposing accruals from goods received not invoiced	Classification with rules	Proposal only; service-date discipline is human
Monthly	Variance narrative	Drafting	Speed, with every figure recomputed
Monthly	Cost at completion candidates and early warning	Forecasting from tabular data	Genuine early warning; see `AIG-04`
Monthly	Schedule health and logic checking	Anomaly detection, rules	Very high value; see `AIG-05`
Monthly	Progress-based slip prediction	Forecasting from tabular data	A second opinion on the critical path analysis, not a replacement
Commercial	Portfolio sweep for notification windows and claim triggers	Extraction, RAG	Coverage no team can match; entitlement stays human
Risk	Candidate risks from analogous history and correspondence	Extraction, pattern detection	Fills gaps in a register; cannot see novel risk
Risk	Quantification and simulation	Simulation support	Runs the model; the inputs and the chosen confidence level are yours
Reporting	Assembling the pack, answering questions over it	RAG, drafting, BI querying	Fast, and dependent on one governed definition per metric
Closeout	Extracting lessons from records	Extraction, summarisation	Good at surfacing candidates; the lesson is a judgement

Nothing in that table changes who signs. The shape is constant: **governed input → AI step → the professional's verification → an output that a named person owns.**

4. Where the capability is weak

This is the half of the map that matters most, because expectations set here determine whether an initiative is judged a success.

Causal reasoning. A model learns association. It will report that cost rose when a particular subcontractor was on site; it cannot tell you that the subcontractor caused it, or that both followed from a late design release. In controls, cause is the deliverable — a variance narrative that misattributes is worse than no narrative, because it directs the recovery action at the wrong thing.

Novel scope. Learning from history requires comparable history. First-of-a-kind work, a new contracting model, an unfamiliar jurisdiction or a technology with no delivery record gives a model nothing to learn from, and models rarely announce that they are extrapolating. The professional's judgement about comparability is the control here, and it cannot be automated.

Contractual judgement. Extraction finds the words; entitlement is a legal question about facts, notices and conduct. A model can tell you that a clause exists and where; it cannot tell you whether you are entitled under it. Entitlement-bearing extractions go to a qualified reviewer before they move a commercial position.

Arithmetic across several steps. Language models predict plausible continuations, and a chain of calculation is exactly where plausible and correct come apart. Arithmetic belongs in a spreadsheet, a script or a calculation engine — with the model used to draft or explain the working, not to be it.

Change in the world. A model learned a portfolio mix, a market and a coding structure. When any of those shift, performance degrades quietly rather than failing loudly. Monitoring is not optional for anything that feeds a decision; `AIG-09 — Bias, explainability and auditability` covers what to monitor and how to evidence it.

Accountability. Unchanged and unchangeable: a model cannot be questioned, cannot be sanctioned and cannot sign. Every output remains explainable, validated and owned by a competent human.

5. Rule, model or generative tool — a triage

The most expensive error on this map is reaching for a model where a rule belongs. Rules are transparent, testable, cheap to audit and do not drift. Use them wherever the logic is known.

- **Use a rule** when the decision is deterministic and stateable: an invoice whose price differs from the purchase order beyond tolerance; an activity with no successor; a posting to a closed period; a change above a delegated authority. If you can write it down, write it down.
- **Use machine learning** when there is a pattern in data worth learning that you cannot state as a rule: which cost lines are likely duplicates given vendor, amount and timing; which activities historically slip; which projects overrun given their features.
- **Use a generative tool** when the task is producing or transforming language: drafting, summarising, extracting from documents, explaining. Ground it in your documents where the answer must come from them, and verify.
- **Use nothing** when the data is inadequate, when confidentiality cannot be assured, when a rule already works, or when the verification the stakes demand would cost more than the task. Deciding not to use AI is a professional judgement, and functions that never make it are not exercising one.

6. How this goes wrong

A general assistant is asked to do precise arithmetic. It produces a beautifully formatted table of figures, some of which are wrong. The class was wrong: this was a spreadsheet task with a drafting layer on top, not a drafting task.

A document question is asked without grounding. Without retrieval over your actual contracts, the model answers from general patterns and invents a plausible clause reference. The failure looks like a correct answer, which is why ungrounded extraction is the most dangerous single habit in this list.

The pilot is chosen for visibility rather than fit. Executive dashboards and natural-language querying demonstrate well and depend on the cleanest data and the most contested definitions. Coding and reconciliation demonstrate poorly and work. Pilot where the class fits and the data exists.

Thresholds are set to make the exception queue small. An anomaly detector tuned until it stops complaining has been tuned into uselessness. The queue size is a consequence of the threshold, and the threshold is a control decision with an owner, not a comfort setting.

Only the hit rate is measured. A detector that is right about 90 % of what it flags may still be missing half of what is there. Both numbers are needed before anyone decides the manual check can stop.

Capability is assumed to transfer. A model that codes cost well on one portfolio is deployed to another with a different coding structure and a different vendor population, and nobody re-tests. Class fit is general; performance is local.

The professional's step is described but not staffed. Every row of §2 has a "what the professional still owes" column. If nobody's time is allocated to that column, the map has been read as an automation plan rather than as a division of labour.

7. Worked example — what coverage is worth

Illustrative figures. A commercial team holds 48 live subcontracts. Each carries roughly 140 pages of contract and, over the year, some 220 pages of correspondence, requests for information (RFIs) and notices — the material in which a missed notification window hides.

`Total pages = 48 × (140 + 220) = 48 × 360 = 17,280 pages`

At a sustained review rate of 30 pages an hour — a generous assumption for careful reading — full human coverage costs:

`17,280 ÷ 30 = 576 hours`, which at 8 hours a day is `576 ÷ 8 = 72 person-days`

No commercial team of ordinary size reads 17,280 pages looking for something that may not be there. Sampling is what actually happens, and sampling means the missed window is found late or not at all.

An extraction-and-retrieval workflow reads all of it and returns, say, nine subcontracts with delay-notice language and an approaching window. The professional then reads nine, opens each cited clause, confirms each date against the contract, and refers the two with material exposure for legal review.

Read it honestly. The saving is not 72 person-days, because the 72 days were never being spent. What changed is that a control which was previously infeasible is now feasible — the class of capability that matters here is coverage, not cleverness. The correct value argument is the exposure avoided by not missing a window, which is specific to your contracts and should be argued with your own numbers, not asserted. And the workflow only works if the extraction is grounded in the documents and cited; an ungrounded sweep returns nine confident answers that may include a clause that does not exist.

8. Checklist — triage a task in five questions

Use this before a tool is chosen, not after.

1. **Is the logic stateable?** If yes, write a rule and stop. Rules beat models on transparency, auditability and cost every time they are available.
2. **What class is this?** Name it from §2. If the answer is "a bit of everything", the task is not yet defined tightly enough to automate or to test.
3. **Does the data exist, and is it fit?** Which system, which fields, how far back, how clean. If this cannot be answered in a sentence, go to `AIG-03 — Data readiness` before going to a vendor.
4. **What does the professional still owe?** Name the verification step, the person and the frequency. Write it into someone's workload before the pilot starts.
5. **How will we know it is still working in six months?** Name the measure, the sample and the cadence. A capability with no monitoring plan is a capability with an unknown current performance.

Related

- `AIG-01 — AI in project controls — the executive guide` — the approval requirements this map supports.
- `AIG-03 — Data readiness: what AI needs before it is any use` — what question 3 of the checklist actually requires.
- `AIG-04 — AI-assisted cost forecasting` — the forecasting class, worked through in method detail.
- `AIG-05 — AI in scheduling — and what must not be automated` — the anomaly-detection and forecasting classes applied to the schedule.
- `AIG-11 — Evaluating AI tools — a buyer's due-diligence guide` — testing a specific product once the class is chosen.

Sources and standards

- **PCI Body of Knowledge, Domain 13** — AI for project controls and project management (Institute manuscript, 2026), particularly its tool-category map and its account of strengths and hard limits, which this document restates for practitioners in capability-class terms.

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AIG-03 · AI IN PROJECT CONTROLS GUIDE

Data readiness: what AI needs before it is any use

Coding structures, cut-off discipline, master data and labels — and what to do when they are not there

VERSION 1.0 · DRAFT
2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

Summary

Most AI failures in a project controls function are data failures wearing a model's clothes. This document sets out what AI actually needs from controls data — a coding structure that is used rather than merely published, cut-off discipline, governed master data, comparable and normalised history, and above all usable labels — and gives an eight-question readiness test naming the artefact, the field, the frequency and the owner. It then deals with the situation practitioners are actually in: what to attempt when the data is not ready and will not be ready this year.

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Contents

Data readiness: what AI needs before it is any use

In one paragraph. Most AI failures in a project controls function are data failures wearing a model's clothes. This document sets out what AI actually needs from controls data — a coding structure that is used rather than merely published, cut-off discipline, governed master data, comparable and normalised history, and above all usable labels — and gives an eight-question readiness test naming the artefact, the field, the frequency and the owner. It then deals with the situation practitioners are actually in: what to attempt when the data is not ready and will not be ready this year.

Who this is for. Cost engineers, cost managers, planners and project controls managers preparing a function for AI, and the data owners in finance and commercial whose records they depend on.

1. The failure is usually upstream

When an AI initiative in controls disappoints, the post-mortem almost never finds a bad algorithm. It finds a cost ledger in which a fifth of the lines sit in a general or miscellaneous code; accruals posted on invoice date rather than service date, so three months of the history are shifted by a period; two vendor records for the same supplier; a percentage complete that means one thing in the schedule and another in the cost report; and nine completed projects, three of which were restated after closeout, offered as "historical data".

None of that is exotic. It is the ordinary condition of controls data in a busy organisation, and it has always been tolerable because a human reading the ledger silently corrects for it — the cost engineer knows that code 9990 is where the site supervisor puts anything awkward, and mentally re-allocates. A model does not know. It learns the miscoding as if it were the business. This is the practical content of "garbage in, garbage out": not that the data is wrong in obvious ways, but that its known-and-tolerated distortions become the pattern the model treats as truth.

The consequence for planning is direct. **Data remediation is not the preliminary to the AI programme; for the first year it usually is the AI programme.** A function that budgets for licences and not for coding work has mispriced the initiative.

2. What AI actually needs

Six things, in the order they usually bite.

2.1 A coding structure that is used, not merely published

The work breakdown structure (WBS), cost breakdown structure (CBS) and code of accounts are what make a cost line comparable to another cost line. Their design belongs to `BPG-03 — Cost` `breakdown structure and the code of accounts`; what matters here is a narrower question: **is the structure actually applied?**

Three diagnostics, each answerable from your own ledger this afternoon. First, the **catch-all share**: what proportion of value and of lines sits in general, sundry, miscellaneous or default codes? Anything above a few per cent means the structure is being routed around. Second, the **obsolete-code share**: how many postings use codes retired in an earlier structure, and is there a mapping table from the old structure to the current one, owned by a named person? Third, the **granularity mismatch**: does the level at which cost is coded correspond to the level at which the schedule is planned? A model asked to relate cost to progress across a mismatch will learn noise.

2.2 Cut-off discipline

Cut-off is the discipline that puts a cost in the period in which the work happened. It is covered as practice in `BPG-07 — Accruals and cut-off discipline`; its relevance here is that cut-off errors are **systematic**, and systematic errors are exactly what models learn most eagerly.

The specific failures that matter: accruals raised from document date rather than service date; a month closed before late invoices arrive, with no accrual, so the cost appears in the following period; period locking that is not enforced, so a prior period changes after a model has been trained on it; and restatements that are made in the reporting layer but not in the source, so the ledger and the reported history disagree. If your monthly actuals move after the fact, record when and by how much — a model that cannot tell a real trend from a posting-lag artefact will forecast the artefact.

2.3 Governed master data

Master data is the reference information every transaction points at: vendors, cost elements, resources, calendars, currencies, units of measure, contract records. It fails in ordinary ways — the same supplier under three spellings, a unit of measure recorded as "m2" in one system and "sqm" in another, two calendars with different holiday sets, currency held without the rate basis used.

The control is unglamorous and specific: **one owner per master-data domain**, a documented process for creating and retiring records, a periodic duplicate check, and a rule about what happens to historical records when a master record is merged. Duplicate-detection and anomaly work is worthless where the vendor master itself is duplicated: the model will faithfully report that two different suppliers submitted identical invoices.

2.4 Labels — the thing you are asking the model to predict

This is the point at which most controls datasets turn out to be far smaller than they look, and it is the single most under-discussed item on the list.

Supervised learning — the kind that predicts a cost at completion, a probability of overrun, an actual finish date — learns from **examples with known answers**. The known answer is the label. In controls the label is usually a project-level or control-account-level outcome: what the final cost actually was, when the work actually finished, whether the risk actually occurred. A ledger with hundreds of thousands of transaction rows may contain a few dozen such outcomes, and fewer that can be trusted.

Three questions establish whether you have labels at all. **Is the outcome recorded in a field, or only in a closeout report?** A number in a document is not a label. **Is the recorded outcome final?** A final cost restated after closeout, or an actual finish taken from a schedule never updated after handover, is a mislabelled example, and mislabelled examples do more damage than missing ones. **Is the outcome attributable to the same unit as the input?** A final cost at project level cannot be learned from features recorded at package level unless someone has done the mapping.

2.5 Comparable history, normalised

Historical projects are only informative if the comparison is fair. Before history is fed to anything, four normalisations are usually needed, and each should be a stated, versioned assumption rather than a habit: a **price basis** (costs brought to a common date with a stated escalation basis), a **location basis**, a **scope basis** (what was in and out — a project that self-performed what another subcontracted is not comparable without adjustment) and a **contract basis** (a reimbursable job and a lump-sum job have different cost behaviour).

Skip normalisation and the model learns escalation and calls it productivity.

2.6 One definition per number, and lineage

Two controls that cost little and prevent a great deal. **One governed definition per metric** — percentage complete, committed cost, cost to date, forecast — held centrally and used by every report and every tool. Where two systems compute the same metric differently, decide which is authoritative and record the reconciling difference rather than letting both circulate. **Lineage** is the ability to trace a figure to its source and through every transformation. It is what allows an AI-influenced number to be defended months later, and its absence is what turns a challenged number into an argument about memory. `AIG-09 — Bias, explainability and auditability` covers the evidencing standard.

3. The readiness test

Eight questions. Each has an artefact, an owner and a cadence, because "we should improve data quality" is not an action.

#	Question	Artefact that answers it	Owner	Cadence
1	What share of cost lines and value sits in catch-all codes?	Ledger extract profiled by code	Cost manager	Monthly
2	Do postings use retired codes, and is there a current mapping table?	Code-of-accounts mapping register	Cost manager	On structure change
3	Are accruals raised on service date?	Accrual listing with service and document dates	Financial controller	Monthly at close
4	Do closed periods stay closed, and are restatements logged?	Period-lock report and restatement log	Financial controller	Monthly
5	How many duplicate master records exist in vendor, resource and cost element?	Master-data duplicate report	Master-data owner	Quarterly
6	How many completed units have a recorded, final, attributable outcome?	Label inventory — one row per completed project or control account	Controls manager	Quarterly
7	Is history normalised to a stated price, location, scope and contract basis?	Normalisation basis note, versioned	Estimating lead	On each dataset build
8	Is there one governed definition per reported metric, with lineage to source?	Metric definitions register and data-lineage map	Controls manager	Half-yearly

A function that can produce all eight artefacts is ready to pilot with confidence. A function that can produce none of them should not conclude that AI is unavailable to it — see §4 — but should certainly not begin with a predictive model.

4. When the ideal is unavailable, which is usually

Very few functions pass §3 on the first attempt, and waiting for a data programme to finish is a way of never starting. Four honest routes forward.

Start where history is not required. Extraction, retrieval-grounded answering over documents, classification against current rules and schedule logic checking need no historical labels. They work on today's material. This is why document-heavy and rules-heavy tasks are the sensible first pilots for a function with poor history — the value arrives while the ledger is still being cleaned.

Use rules while the data matures. Deterministic checks — tolerance breaches, missing successors, postings to closed periods, approvals out of sequence — are transparent, auditable and immune to the data problems that defeat models. They also generate the exception discipline a model will later need.

Build the label set forwards. If there is no reliable record of final cost, actual finish and realised risk, start recording one now: a single register, one row per completed control account, populated at closeout by a named person as a condition of closeout. Two years of deliberate labelling is worth more than ten years of archaeology.

Narrow the scope until the data supports it. A model over one repeatable package type with fifty clean instances will outperform a portfolio-wide ambition with nine mixed ones. Narrow, prove, extend.

What is not an acceptable route is training on data known to be distorted and treating the output as indicative. An indicative number that is wrong in a consistent direction is more dangerous than no number, because it will be quoted.

5. How this goes wrong

Volume is mistaken for readiness. "We have ten years of data" describes a row count. The question is how many labelled, comparable, trustworthy examples exist, and the answer is usually one or two orders of magnitude smaller. §6 works this through.

The data assessment is done by the vendor. A supplier profiling your data to establish whether their product will work is not the same as your function establishing whether its data supports the decision. Both may be worth doing; only one is assurance.

Cleaning is done once, for the pilot. A one-off remediation produces a clean training set and a dirty production feed. Within two cycles the live data no longer resembles what the model learned, and performance degrades without anyone deciding anything.

The catch-all code is fixed by renaming it. Splitting "miscellaneous" into six specific codes achieves nothing if the site team still has no time to code accurately. The remedy is at the point of capture — default coding rules, a shorter picklist, a purchase-order-driven default — not in the chart of accounts.

Confidential data is prepared carelessly. Anonymisation that leaves project names in free-text fields, or a "cleaned" extract that retains personal data in a comment column, is a confidentiality incident waiting to happen. Preparation includes checking what left with the data, not only what was intended to.

Restatements are silently absorbed. Prior-period figures change and nothing records that they did, so the difference between a real trend and a posting correction becomes unrecoverable. A restatement log costs one line per event and saves an entire class of forecasting error.

Definitions drift between the tool and the report. A natural-language query answers from a subtly different definition of committed cost than the controlled report uses, and two credible numbers now circulate. Reconcile every AI-produced figure to the governed one before it is quoted, and fix the definition rather than the answer.

6. Worked example — a readiness profile

Illustrative figures. A controls function proposes to build a cost-at-completion model. It offers 24 months of cost ledger across nine completed projects: **26,400 rows**. Before anything is built, the data is profiled against the checks of §3.

Check	Rows failing	Share of 26,400
Invalid or retired cost code	1,320	`1,320 ÷ 26,400 = 5.0 %`
Missing or defaulted WBS reference	792	`792 ÷ 26,400 = 3.0 %`
Posted in a period after the service date crossed a cut-off	1,584	`1,584 ÷ 26,400 = 6.0 %`
Duplicate candidate (same vendor, amount and date)	264	`264 ÷ 26,400 = 1.0 %`
Currency or unit-of-measure inconsistency	396	`396 ÷ 26,400 = 1.5 %`
Total flags	**4,356**	`4,356 ÷ 26,400 = 16.5 %`

Flags are not rows: some rows fail more than one check. De-duplicating the flag list gives **3,696 distinct rows affected**, which is `3,696 ÷ 26,400 = 14.0 %` of the dataset, with `4,356 − 3,696 = 660` rows counted twice.

Remediation effort, at an assumed 2 minutes per affected row for review and re-coding:

`3,696 × 2 = 7,392 minutes = 7,392 ÷ 60 = 123.2 hours = 123.2 ÷ 8 = 15.4 person-days`

Then the sentence that matters. The model is to predict final cost. The label is one number per completed project, so the dataset does not have 26,400 examples; it has **nine**. Of those nine, three were restated after closeout and the restatement was never carried back to the source ledger, so three labels are known to be wrong. **Six trustworthy labelled examples.**

Result. Do not build the predictive model. Do the 15.4 days of remediation because it improves this month's reporting regardless; begin a label register populated at closeout; and put the AI effort into extraction and rules-based checking, which need no labels. Revisit the model when the label count supports it.

Assumptions this answer depends on. That 2 minutes per row is a measured rate for this team, not an estimate; that the five checks are independent enough for the overlap figure to be meaningful; and that project-level final cost is the right prediction target at all — a control-account-level target would give more labels, at the price of needing control-account-level features.

7. Checklist — before you feed anything to a model

1. **Profile first, build second.** Run the five checks of §6 on the actual extract, and record the result with a date. Do not accept a description of the data in place of a profile of it.
2. **Count labels, not rows.** State how many completed units have a final, attributable, unrestated outcome. If the number is in single figures, the honest answer is that you do not have a training set.
3. **Name the price, location, scope and contract basis** to which history has been normalised, and version the note.
4. **Confirm the cut-off rule** — service date, not document date — and confirm that closed periods are locked and restatements logged.
5. **Check the master data** for duplicates in vendor, resource, cost element and calendar before relying on any anomaly or duplicate detection.
6. **Fix one definition per metric** and reconcile every AI-produced figure to the governed one.
7. **Confirm what left with the extract**: personal data, project names in free text, commercially sensitive rates. Preparation is a confidentiality control, not only a quality one.

8. **Write down what you will not do yet**, and what would change that. A readiness assessment that concludes "not yet" has done its job.

Related

- `AIG-02 — What AI actually does in a controls function` — which capability classes need history and which do not, the distinction §4 turns on.
- `AIG-04 — AI-assisted cost forecasting` — the first place inadequate labels and unnormalised history do visible damage.
- `AIG-09 — Bias, explainability and auditability` — the evidencing and monitoring standard that lineage supports.
- `BPG-03 — Cost breakdown structure and the code of accounts` — designing the structure whose use §2.1 tests.
- `BPG-07 — Accruals and cut-off discipline` — the practice behind §2.2.

Sources and standards

- **PCI Body of Knowledge, Domain 13** — AI for project controls and project management (Institute manuscript, 2026), Knowledge Area 13.2 on data quality dimensions, structure, governance and lineage, which this document extends into a readiness test and a not-ready-yet route.

Status and version

Founding-stage document · Version 1.0 · draft · Reviewed under PCI governance. PCI makes no claims of accreditation or recognition beyond what is true today.

AIG-04 · AI IN PROJECT CONTROLS GUIDE

AI-assisted cost forecasting

What a model may contribute to a forecast, what it must never decide, and how to validate a number it produced

VERSION 1.0 · DRAFT

2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

Summary

A method document on using AI in cost forecasting. It sets out the six contributions a model can legitimately make — completeness of actuals, candidate estimates at completion, driver decomposition, portfolio trend detection, ranges and narrative drafting — and the five decisions that remain human, including which forecast to defend, whether a variance cause is closed, baseline change and contingency release. Its centre is a seven-step validation protocol for an AI-produced forecast number, worked through on a control account with the substitutions shown, so a practitioner can apply it at the next month-end.

About this document

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Contents

AI-assisted cost forecasting

produced.

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Who this is for. Cost engineers, cost managers, control account managers and project controls managers who produce or challenge a forecast, and the project managers who have to defend one.

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1. The problem a forecast actually has

A forecast is a claim about the future that a named person must defend, sometimes years later, sometimes in a dispute. Its quality has never rested mainly on the arithmetic — the arithmetic of the earned value management (EVM) family is simple — but on three harder things: whether the cost to date it starts from is complete, whether the method chosen matches the reason the variance occurred, and whether the person presenting it can explain the movement since last period in terms a director can act on.

AI changes the cost of producing candidates and explanations. It does not touch the defence. A model that returns an estimate at completion (EAC) in under a second has given you a number and none of the three things above.

The terms used throughout, expanded once: **planned value (PV)** is the budgeted cost of work scheduled; **earned value (EV)** the budgeted cost of work performed; **actual cost (AC)** the cost incurred for that work; **budget at completion (BAC)** the total authorised budget; **estimate at completion (EAC)** the forecast final cost; **estimate to complete (ETC)** the forecast cost of the remaining work; **cost performance index (CPI)** is `EV ÷ AC`; **schedule performance index (SPI)** is `EV ÷ PV`; **to-complete performance index (TCPI)** is the efficiency the remaining work must achieve to hit a target; **variance at completion (VAC)** is `BAC − EAC`. Method selection among the EAC family belongs to `BPG-09 — Estimate at completion`; this document assumes it and concentrates on the AI contribution and the validation.

2. What AI can legitimately contribute

2.1 Completeness of the cost to date. The most common cause of a wrong forecast is not a wrong method but an incomplete AC: goods received not invoiced (GRNI) unaccrued, commitments not reflected, subcontractor valuations in transit, a late invoice run. Classification and matching capability is genuinely good at reconciling purchase orders, receipts and invoices at volume, proposing accruals from receipt records, and flagging the lines where the three disagree. This is the highest-value, lowest-risk contribution on the list, and it improves the forecast without touching the method at all.

2.2 Candidate EACs, generated across methods. Producing the CPI-based, composite and remaining-work-at-budget variants for every control account, every period, is tedious and therefore is often done for a handful of accounts. A model does all of them and ranks the divergence. The value is not the numbers — they are arithmetic — but the **spread**: accounts where the methods agree need less attention than those where they diverge widely, which is a useful way to direct scarce analytical time.

2.3 Driver decomposition. Given cost, quantity, rate and commitment data, a model can decompose a movement into components: rate against usage, escalation against scope, this package against that one. Decomposition is arithmetic and it is checkable. **Attribution of cause is not decomposition** — the model can tell you that 130,000 of the movement is in rework quantities; it cannot tell you that the rework was caused by late design, and it should not be asked to.

2.4 Trend and early warning across a portfolio. Watching CPI, quantity growth, commitment burn and productivity across hundreds of accounts, and raising a signal when a trend is sustained rather than when a threshold is breached, is work that scales badly for humans and well for machines. Sustained-decline detection across three or four periods is the single most useful early-warning pattern in cost control.

2.5 Ranges and scenarios. Producing a range around a central forecast, and running scenarios — escalation at a different rate, productivity recovering or not — lets the professional present a range with named drivers instead of a single point with false precision. The confidence level and the scenario set are professional choices; the computation is not.

2.6 Narrative drafting. A first-draft variance narrative from the numbers, in the house format, saves real time. Every figure in it is recomputed before it leaves, and every causal claim is either confirmed against the actual analysis or deleted.

3. What a model must not decide

Five decisions sit outside the model regardless of how well it performs. `AIG-10 — Human in the loop` owns the general rule; these are the specifics for cost forecasting.

Which EAC is reported. Selecting the forecast to be defended is a judgement about the future behaviour of a specific job. The model proposes candidates; the professional selects, states the assumption the selection embodies, and owns it.

Whether the variance cause is closed or persisting. This is the assumption on which the entire EAC family turns — the CPI-based EAC assumes the past performance continues, the remaining-work-at-budget variant assumes it does not. Answering it requires knowing whether the design error was corrected, whether the crew has been replaced, whether the supplier has been changed. That knowledge is on the job, not in the data.

Baseline change. A forecast that exceeds budget is a forecast; a change to the budget is a governance act with an approver and an audit trail. Models must never write to the baseline, and no automated workflow should update a budget field. `BPG-04 — Baselining and baseline change control` covers the control.

Contingency release. Drawing contingency is a management decision made against defined criteria and appetite, and it is the point at which a forecast becomes money. A model may compute the exposure and the remaining balance; the release decision is made by the named authority, minuted.

Reclassification of scope as variation. Whether work is a variation, a claim or absorbed scope is a contractual judgement with entitlement consequences, and it moves the forecast materially. Extraction may locate the clause and the correspondence; the position is taken by a commercial professional, with legal review where the exposure warrants it.

4. Validating an AI-produced forecast number

Seven checks, in order. The first four are non-negotiable for any figure that will be reported; the last three are proportionate to the stakes.

1. **Recompute from source.** Take PV, EV, AC and BAC from the controlled system, not from the model's output, and recompute CPI, SPI and the candidate EACs by hand or in a controlled sheet. If the model's inputs and yours differ, stop: the discrepancy is the finding.
2. **Check the method against the cause.** Name the assumption the model's EAC embodies — performance continues, performance recovers, remaining work at budget — and test it against what is actually happening on the job. A model with no site knowledge will usually assume continuation or a history-derived recovery, and the second of those is where quiet optimism enters.
3. **Run the TCPI reality check.** Compute the efficiency the remaining work must achieve to land on the budget, and compare it with the efficiency achieved so far. A required index far above the achieved one is a forecast that has not yet admitted what it knows. Note the trap in §6: TCPI computed against a CPI-based EAC returns the achieved CPI by construction and checks nothing.
4. **Reconcile to commitments and the cost to date.** Confirm that the AC used is complete — accruals on service date, GRNI included, subcontract valuations current — and that the forecast is not lower than the sum of committed cost plus reasonable remaining exposure.
5. **Bridge the movement.** Explain the change from last period as a small number of named components that sum exactly to the movement. A movement that cannot be bridged is not yet understood, whoever produced it.
6. **Compare against one independent simple method.** A quantity-based or rate-based estimate to complete, done crudely, is a valuable check on a sophisticated one. Where the two disagree materially, the sophistication is on trial, not the simplicity.
7. **Check for double counting.** Risk allowances carried both in the forecast and in contingency; escalation applied both in the rate and in a provision; a variation counted in both the forecast and the change log. Double counting is the most common defect in AI-assembled forecasts, because assembly from multiple sources is exactly what the tooling makes easy.

Where the model was trained on your historical projects, one further check applies before reliance rather than at each use: **is the training history comparable to this job** in scope, contract type, location and price basis? `AIG-03 — Data readiness` §2.5 gives the normalisation this depends on.

5. How this goes wrong

The model's number becomes the anchor. The first figure on the page shapes every discussion that follows. When a model produces the first figure, the professional's role silently changes from forecasting to adjudicating the model — and adjudicating downwards is uncomfortable in a meeting. Produce your own candidate before you look at the model's, at least on material accounts.

Recovery is assumed because history recovered. A model trained on completed projects learns that late-project CPI often improves — partly because projects genuinely recover, and partly because late scope gets moved, claims get settled and budgets get changed. It cannot distinguish those, so it forecasts recovery on a job where nothing has changed.

The forecast is precise and unfounded. An EAC quoted to the currency unit from inputs that are ranges communicates confidence the analysis does not have. Round to the precision the inputs support and state the range.

Trend detection is tuned until it stops complaining. Early warning that generates too many signals gets turned down until it generates none, and the tuning is done by whoever finds the alerts annoying rather than by whoever owns the risk. Threshold changes are control changes: they get an owner and a record.

The narrative is believed because it is fluent. A drafted variance narrative attributes the overrun to weather; the actual analysis shows a rate variance on imported material. Fluency is not evidence. Every causal claim in a generated narrative is either traced to the analysis or removed.

Incomplete actuals are forecast forward. The most expensive arithmetic error in cost control is forecasting from an AC missing a month of accruals: CPI looks strong, the EAC looks comfortable, and both correct sharply when the invoices land. Accelerating the cycle makes this worse, because there is less elapsed time for late documents to arrive before the pack closes.

Nobody re-tests the model after the portfolio changes. A model that performed well on building work is applied to civils, or the delivery model shifts from reimbursable to lump sum, and performance degrades without an alarm. Re-test on a change in the work, not only on a calendar.

6. Worked example — validating an AI-produced EAC

Illustrative figures. One control account, month 14. Currency USD, all figures cumulative to the data date, rounded to the nearest whole currency unit. Not benchmark data.

Controlled inputs, taken from source rather than from the model:

Measure	Value
Budget at completion (BAC)	8,400,000
Planned value (PV)	5,000,000
Earned value (EV)	4,620,000
Actual cost (AC)	5,100,000

Step 1 — recompute.

`CV = EV − AC = 4,620,000 − 5,100,000 = −480,000` `SV = EV − PV = 4,620,000 − 5,000,000 = −380,000` `CPI = EV ÷ AC = 4,620,000 ÷ 5,100,000 = 0.906` (0.905882, shown to three decimals) `SPI = EV ÷ PV = 4,620,000 ÷ 5,000,000 = 0.924`

Step 2 — candidate forecasts.

Performance continues, cost only: `EAC = BAC ÷ CPI = 8,400,000 ÷ 0.905882 = 9,272,727` (equivalently `BAC × AC ÷ EV = 8,400,000 × 5,100,000 ÷ 4,620,000 = 9,272,727`)

`VAC = BAC − EAC = 8,400,000 − 9,272,727 = −872,727`

Performance continues, cost and schedule pressure both persisting: `CPI × SPI = 0.905882 × 0.924 = 0.837035` `ETC = (BAC − EV) ÷ (CPI × SPI) = 3,780,000 ÷ 0.837035 = 4,515,939` `EAC = AC + ETC = 5,100,000 + 4,515,939 = 9,615,939`

The model's proposal: EAC = 8,960,000, with a decomposition attributing the variance largely to a design-rework driver and a note that comparable accounts in the training history recovered part of their cost variance after the rework driver closed.

Step 3 — TCPI reality check. To land on budget, the remaining work must be performed at:

`TCPI to BAC = (BAC − EV) ÷ (BAC − AC) = 3,780,000 ÷ (8,400,000 − 5,100,000) = 3,780,000 ÷ 3,300,000 = 1.145`

Achieved efficiency is 0.906. Nothing on the account has changed that would support a jump to 1.145, so any forecast at or near budget is already refuted.

The trap. Computing TCPI against the CPI-based EAC gives `(BAC − EV) ÷ (EAC − AC) = 3,780,000 ÷ (9,272,727 − 5,100,000) = 3,780,000 ÷ 4,172,727 = 0.906` — exactly the achieved CPI, by construction. It looks like a passed check and is arithmetically empty. The check that bites is against the budget, or against a different forecast.

Step 4 — reconcile to commitments. Committed cost on the account is 7,940,000, of which 5,100,000 is incurred. A forecast of 8,960,000 leaves `8,960,000 − 7,940,000 = 1,020,000` for all remaining uncommitted work and for any growth in the committed packages. The professional checks that figure against the remaining scope; if the uncommitted scope is plainly worth more than 1,020,000, the model's forecast fails here, before any argument about method.

Step 5 — bridge the movement. Last period's reported EAC was 9,050,000; this period's CPI-based candidate is 9,272,727, a movement of `9,272,727 − 9,050,000 = +222,727`. The decomposition offered:

`+150,000` rate escalation on imported material · `+130,000` rework quantities in the containment package · `−57,273` scope removed by approved variation

`150,000 + 130,000 − 57,273 = 222,727` — the bridge closes exactly, and each component is then traced: the escalation to the procurement record, the rework to the non-conformance log, the variation to the change register.

Step 6 — the judgement. The model's 8,960,000 sits below both candidates because it assumes partial recovery once the rework driver closes. The professional's question is not whether the model is sophisticated but whether **this** driver has closed on **this** account. If the design revision is issued, the rework scope is bounded and the crew is back on planned work, a recovery assumption is defensible and should be stated as an assumption. If the revision is still in review, it is not, and the honest forecast sits between the CPI-based 9,272,727 and the composite 9,615,939.

Step 7 — what is reported. A forecast of **9,270,000** (rounded to the nearest 10,000, reflecting input precision), stated as: performance-continues basis, cost only; assumes the rework driver closes in the current period with no further growth; range to 9,620,000 if schedule pressure persists; TCPI to budget of 1.145 shown as the reason budget is no longer credible; AI-assisted, verified and owned by the named cost engineer.

Assumptions this answer depends on. That EV is measured on rules of credit that have not changed this period; that AC is complete at the data date, including accruals on service date; that the approved variation is fully reflected in both BAC and the change register; and that the training history behind the model's recovery assumption is comparable to this account, which §4's comparability check must establish separately.

7. Checklist — before an AI-assisted forecast leaves the function

1. **Inputs from source.** PV, EV, AC and BAC taken from the controlled system and recomputed, not accepted from the model's output.
2. **Cost to date complete.** Accruals on service date, GRNI included, subcontract valuations current, commitments reconciled.
3. **Assumption named.** The forecast's assumption about the variance cause is stated in one sentence a director can challenge.
4. **TCPI to budget computed** and shown, not TCPI to the forecast that produced it.
5. **Movement bridged** to components that sum exactly, each traced to a record.
6. **Independent check done** by one simple method, with any material disagreement explained.
7. **No double counting** between forecast, risk allowance, contingency and change log.
8. **Precision honest.** Rounded to what the inputs support, with a range where one exists.
9. **Ownership recorded.** Named professional, AI assistance disclosed, and a note of what the model proposed and what changed in review.

Related

- `AIG-03 — Data readiness: what AI needs before it is any use` — the comparability and cut-off conditions §4 and §5 depend on.
- `AIG-05 — AI in scheduling — and what must not be automated` — the schedule side of the same forecast, and why a cost forecast that ignores the critical path is incomplete.
- `AIG-10 — Human in the loop: what AI may and may not decide` — the general decision-rights schedule behind §3.
- `BPG-08 — Earned value in practice` — the measurement discipline the inputs to §6 assume.
- `BPG-09 — Estimate at completion — choosing and defending a method` — how the method is selected, which this document treats as given.

Sources and standards

- **PCI Body of Knowledge, Domain 6** — Earned value management and forecasting (Institute manuscript, 2026). The forecasting family and the TCPI reality check are explained there; this document applies them to an AI-produced number.
- **PCI Body of Knowledge, Domain 13** — AI for project controls and project management (Institute manuscript, 2026), Knowledge Area 13.5.3, the forecasting workflow whose propose–verify–own shape §4 operationalises.

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AIG-05 · AI IN PROJECT CONTROLS GUIDE

AI in scheduling — and what must not be automated

Where machine checking earns its place in a planning function, and the decisions that stay with the planner

VERSION 1.0 · DRAFT

2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

Summary

A method document on AI in planning and scheduling. It sets out what machine capability genuinely contributes — logic and health checking at volume, duration realism against history, progress-based slip prediction, scenario assembly and integration checks against cost — and the eight decisions that must never be automated, from accepting a critical path to inferring progress, fixing dates with constraints and judging delay causation. Its centre is a validation protocol for an AI-produced schedule output and a worked example in which a health sweep uncovers a hidden constraint concealing a slip.

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Contents

AI in scheduling — and what must not be automated

planner.

In one paragraph. A method document on AI in planning and scheduling. It sets out what machine capability genuinely contributes — logic and health checking at volume, duration realism against history, progress-based slip prediction, scenario assembly and integration checks against cost — and the eight decisions that must never be automated, from accepting a critical path to inferring progress, fixing dates with constraints and judging delay causation. Its centre is a validation protocol for an AI-produced schedule output and a worked example in which a health sweep uncovers a hidden constraint concealing a slip.

Who this is for. Planners, planning managers, project controls managers and the commercial staff who rely on a programme as a contractual record.

1. What a schedule is, and why that constrains automation

A schedule is not a forecast of dates. It is an **argument about sequence** — that this work can follow that work, in this duration, with these resources — from which dates are a consequence. It is also, on most contracts, a record with commercial weight: the baseline against which delay is measured, the artefact an extension-of-time (EOT) claim is built on, and often a submission the other party may accept or reject.

That dual character sets the boundary for automation cleanly. Machine capability is excellent at **interrogating the argument** — finding the places where the network cannot mean what it appears to mean. It is not competent to **make the argument**, because the argument rests on knowledge of the work, the site, the resources and the contract that is not in the file. And it must never be allowed to quietly change the record.

Two terms are used throughout: the **critical path method (CPM)** is the network technique that derives dates and float from activity durations and logic; **float** is the time an activity can move without affecting a later constraint or the finish.

2. What AI contributes to a planning function

2.1 Logic and health checking at volume. This is the strongest use, and it is close to pure gain. A checker can examine every activity in a large programme for open ends (activities without a predecessor or successor), hard constraints that fix dates irrespective of logic, long lags substituting for real activities, negative float, out-of-sequence progress, misassigned calendars, durations far outside the population, and logic types used in ways that break the network's meaning. The output is an exception list in minutes rather than the two days a manual review of a large programme takes — which in practice means the review happens every update instead of at baseline only.

`BPG-05 — Schedule quality: a practical review` owns the checks themselves; the contribution here is frequency and coverage.

2.2 Duration realism against history. Where a normalised history of comparable activities exists, a model can flag durations that sit far outside what similar work has actually taken. This is a challenge mechanism, not a correction: the flag opens a conversation with the person who owns the duration, whose reasons may be perfectly good.

2.3 Progress-based slip prediction. Extrapolating achieved progress rates onto remaining work gives a finish date derived from behaviour rather than from planned durations. Its value is precisely that it usually disagrees with the CPM date. The disagreement is the finding — see §6 — and it must be explained, not averaged away.

2.4 Scenario assembly. Building and comparing what-if cases — a delayed permit, a resequenced area, an additional crew — is mechanical work that a model can prepare quickly. The scenarios worth running, and the one adopted, are chosen by people.

2.5 Integration checks against cost and commitments. Cross-checking that an activity showing progress has cost against it, that a package with committed cost has activities in the network, and that the schedule's remaining duration is consistent with the cost forecast's remaining work is tedious, valuable and well suited to automation. It also catches the failure mode where the cost report and the programme have drifted into telling different stories.

2.6 Drafting the schedule narrative. A first draft of the update commentary — what moved, what drove it, what the mitigation is — saves time, with every date and float figure recomputed and every causal claim confirmed before it is issued.

2.7 Preparing risk inputs. Proposing three-point ranges for activities from historical spread gives a starting point for quantitative schedule risk analysis. The ranges are then owned by the people who own the work; `AIG-06 — AI for risk identification and quantification` deals with the quantification discipline and `BPG-17` with the analysis itself.

3. What must not be automated

Eight decisions. The general rule belongs to `AIG-10 — Human in the loop`; these are the scheduling specifics.

Accepting the critical path. A tool reports a driving path; a planner establishes whether it is real by tracing the logic and asking whether the sequence makes physical sense. A driving path produced by a constraint, a calendar artefact or a missing link is a reporting artefact, not a plan.

Baseline acceptance. Accepting a baseline is a contractual and governance act with an approver. No automated process may set, replace or amend a baseline, and no model may write to baseline fields.

Inferring progress. Percentage complete, actual start and actual finish are **measurements**, made against rules of credit by someone accountable for the measurement. A model may draw attention to an activity whose reported progress is inconsistent with its cost, its successors or its resource usage; it may not set the value. Inferred progress is the fastest route to a programme that is confidently wrong.

Inserting or adjusting constraints. A constraint is a statement that something outside the network fixes a date. Each one requires a justification recorded against it. Automated date-fixing — including optimisation features that quietly apply constraints to achieve a target — destroys the network's meaning.

Delay causation and EOT entitlement. Which delay caused which effect, whether it is excusable, whether it is compensable, and what notice was given are contractual determinations resting on facts and conduct. Analysis tools assist; entitlement is decided by people, with legal input where the exposure warrants it. `BPG-12 — Claims and extension of time` owns this ground.

Choosing between mitigation and acceleration. Re-sequencing within existing resources and spending money to compress are different decisions with different commercial consequences and, often, different contractual entitlements. A model may cost and compare the options; a person chooses.

Resource levelling that moves dates. Automatic levelling can produce a feasible-looking programme by moving work in ways nobody has agreed. Levelling output is a proposal that must be inspected activity by activity on the affected paths.

Re-logicking to reach an acceptable date. The most damaging automation of all is any process that adjusts logic, durations or calendars until the finish date is the one required. When the date is wrong, the answer is a mitigation plan or a claim, not a network that has been persuaded.

4. Validating an AI-produced schedule output

Seven checks, applied to any machine-produced schedule, health report or predicted date before it informs a decision.

1. **Establish the status data.** Confirm the data date, and confirm that actual dates, remaining durations and percentage complete came from measurement rather than from a tool's default. Confirm how out-of-sequence progress is being handled — whether the calculation retains the original logic or lets progressed work override it — because the two settings can produce materially different finish dates from identical data.
2. **List and justify every constraint.** Not "count them" — list them, with the reason and owner for each. Anything unjustified is removed and the network re-run.
3. **Trace the driving path by hand.** Take the reported critical path and walk it backwards through the logic for a representative stretch. Confirm each link is a real dependency rather than a sequencing preference, and that no constraint or calendar is doing the driving.
4. **Reconcile the predicted finish with the CPM finish.** Where a trend-based prediction differs from the network date, quantify the difference and explain it in terms of durations and achieved rates. Do not average, and do not report the more comfortable of the two.
5. **Check the calendars.** Confirm activity calendars, holiday sets and shift patterns are the ones intended; calendar errors move dates invisibly and survive every other check.
6. **Work the exception list, do not count it.** A health report that says "26 open ends" and is filed has achieved nothing. Each exception is resolved, justified or accepted with a reason, and the network is re-run afterwards — the re-run is where the finding usually appears.
7. **Cross-check against cost and commitments.** Confirm that the programme's remaining work and the cost forecast's remaining work describe the same job; see `AIG-04 — AI-assisted cost forecasting` §4.

5. How this goes wrong

The exception list becomes the deliverable. A weekly automated health report circulates, the counts trend downwards because the easy items get fixed, and nobody notices that the nine hard constraints have been there since baseline. Counting is not checking.

The checker is trusted on what it does not check. Machine checks find structural defects — missing links, constraints, negative float. They do not find a plausible-looking sequence that is physically impossible, a duration that ignores a permit, or an omitted scope. A clean health report says the network is well formed, not that the plan is sound.

Progress is back-filled from cost. Because an integration check flags activities with cost but no progress, someone resolves the exception by updating progress to match the cost. The check was designed to find measurement failures and has instead been used to manufacture measurements.

Predicted dates are reported without their basis. A trend-based finish date reported alongside the CPM date, with no explanation of why they differ, invites the reader to pick one. The planner's job is to say which is credible and why.

Optimisation quietly fixes dates. A scheduling assistant that "improves" a programme by applying constraints or compressing durations produces a schedule that looks achievable and has stopped modelling the work. Every change a tool proposes is inspected before acceptance.

The delay analysis is done by the tool that produced the schedule. Where the same automated process both maintains the programme and analyses the delay, an error in the first is invisible to the second. Keep the analysis independent, particularly where entitlement is in issue.

Confidential programme data leaves the organisation. A contractor's programme, resource loading and sequencing intent are commercially sensitive. Uploading a submitted programme to an ungoverned tool for "a quick check" is a confidentiality incident with contractual consequences.

6. Worked example — the health sweep and the hidden constraint

Illustrative figures. A 1,480-activity contractor programme is submitted for acceptance at month 24. A machine health sweep is run before the planner's review. Working days throughout; a five-day working week is assumed for calendar conversions.

Step 1 — the sweep.

Finding	Count	Note
Open ends (no predecessor or no successor)	26	$26 \div 1{,}480 = 1.8\ \%$ of activities
Hard date constraints	9	Each requires a justification
Lags longer than 10 days	14	Candidate substitutes for real activities
Activities with out-of-sequence progress	63	$63 \div 1{,}480 = 4.3\ \%$
Remaining duration exceeding original duration	31	Progress reported without duration reassessment

Step 2 — working the list. The planner re-logics the 26 open ends, replaces 11 of the 14 long lags with real activities and justifies the other 3, and reviews the 63 out-of-sequence activities against the actual site sequence. Of the 9 constraints, 8 have recorded reasons; the ninth is a "finish on or before" constraint on a commissioning activity, applied at a previous update with no note.

Step 3 — the re-run. With the unjustified constraint removed and the logic corrected, the programme's completion moves from **working day 640 to working day 658**:

`658 − 640 = 18 working days`, which on a five-day week is `18 ÷ 5 × 7 = 25.2`, say **25 calendar days**

The slip was not created by the review. It existed and was being held out of the reported date by a constraint nobody had justified.

Step 4 — the trend prediction, reconciled. The data date is working day 528, so the corrected CPM leaves `658 − 528 = 130` working days remaining. Over the last eight update periods the programme has achieved progress at approximately **0.90 of the planned rate**. Extrapolating that rate onto the remaining work:

`130 ÷ 0.90 = 144 working days remaining`, giving a finish at `528 + 144 = working day 672`

`672 − 658 = 14 working days` beyond the corrected CPM date, and `672 − 640 = 32` working days beyond the submitted date, or `32 ÷ 5 × 7 = 44.8`, say **45 calendar days**.

The two dates are not competing forecasts to be averaged. The CPM date says: if the remaining durations are achieved as planned, the finish is day 658. The trend date says: nothing in the last eight periods suggests the remaining durations will be achieved as planned. The planner's task is to decide which remaining durations are credible, re-duration where they are not, and say so — or to produce a mitigation plan that explains how the planned rate will now be achieved when it has not been.

Step 5 — the commercial consequence, flagged not decided. Assume for the example a liquidated-damages (LD) rate of USD 25,000 per calendar day of delay. The constraint-concealed slip alone represents `25,000 × 25 = USD 625,000` of potential exposure; on the trend-based date, `25,000 × 45 = USD 1,125,000`.

These are exposure figures, not liabilities. Whether damages apply at all depends on entitlement, notices, concurrency and the contract's mechanism — determinations that belong to the commercial team and `BPG-12 — Claims and extension of time`, not to the planner and certainly not to the checker. The purpose of computing them is to make the priority of the finding legible to people who do not read programmes.

Assumptions this answer depends on. A five-day working calendar with no holiday effects in the affected period; that the achieved-rate figure of 0.90 is measured over a period representative of the work remaining; that the removed constraint had no contractual basis, which the planner confirms before removing it; and that the LD rate is taken from the executed contract rather than assumed.

7. Checklist — before an AI-assisted schedule output is relied upon

1. **Data date and status confirmed**, with actual dates and remaining durations from measurement, not defaults.
2. **Out-of-sequence handling known** and its effect on the finish date understood.
3. **Every constraint listed with a justification and an owner**; the unjustified ones removed and the network re-run.

4. **Driving path traced by hand** for a representative stretch, and confirmed to be logic-driven.
5. **Calendars verified** against the intended working pattern and holiday set.
6. **Exception list worked to closure**, each item resolved, justified or accepted with a reason.
7. **Predicted and CPM dates reconciled**, with the difference explained in durations and achieved rates.
8. **Programme reconciled to cost and commitments** for the same remaining scope.
9. **No automated write-back** to baseline, progress, logic or constraints, and the tool's proposals inspected before acceptance.
10. **Named planner owns the issued programme**, with AI assistance recorded and the review changes noted.

Related

- `AIG-04 — AI-assisted cost forecasting` — the cost half of the same forecast, and the integration check at §4.7.
- `AIG-06 — AI for risk identification and quantification` — where the three-point ranges of §2.7 are quantified, and the confidence discipline that governs them.
- `BPG-05 — Schedule quality — a practical review` — the check definitions this document automates.
- `BPG-12 — Claims and extension of time` — the entitlement ground §3 and §6 deliberately stop short of.
- `TPL-14 — Schedule quality review checklist` — the manual instrument behind the machine sweep.

Sources and standards

- **PCI Body of Knowledge, Domain 10** — Project scheduling (Institute manuscript, 2026). The network fundamentals, float and health-check disciplines applied here.
- **PCI Body of Knowledge, Domain 13** — AI for project controls and project management (Institute manuscript, 2026), Knowledge Area 13.5.5, the scheduling workflow and its warning about hidden constraints and unrealistic durations in machine-assisted schedules.

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AIG-06 · AI IN PROJECT CONTROLS GUIDE

AI for risk identification and quantification

What a model can add to a risk process, and how it manufactures confidence when nobody is watching

VERSION 1.0 · DRAFT
2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

Summary

A method document on AI in project risk work. It separates identification — where machine reading of correspondence, history and the estimate genuinely finds candidate risks a workshop misses — from quantification, where the same tooling can manufacture confidence the inputs do not support. It names what a model cannot see, including the blind spots it inherits from your own register; sets out the six ways a simulation flatters itself; states the decisions that remain human, including the confidence level and contingency release; and works an expected-monetary-value and simulation example in which a single unexamined correlation assumption moves the contingency by more than the smallest risk in the register.

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Contents

AI for risk identification and quantification

In one paragraph. A method document on AI in project risk work. It separates identification — where machine reading of correspondence, history and the estimate genuinely finds candidate risks a workshop misses — from quantification, where the same tooling can manufacture confidence the inputs do not support. It names what a model cannot see, including the blind spots it inherits from your own register; sets out the six ways a simulation flatters itself; states the decisions that remain human, including the confidence level and contingency release; and works an expected-monetary-value and simulation example in which a single unexamined correlation assumption moves the contingency by more than the smallest risk in the register.

Who this is for. Risk managers, cost engineers and planners who maintain quantified registers, project controls managers who set contingency recommendations, and the project directors who approve them.

1. Two failures, and only one of them is new

Risk processes fail in two recognisable ways. The first is old: a register that has become an issues list — things that have already happened, written as risks, with mitigation columns describing work already done. Nothing in it looks forward, and the workshop that produced it drew on the same six people's recent experience.

The second failure is newer and more dangerous: a register that is thin, unexamined and unevidenced, put through a simulation, and returned as a smooth distribution with a contingency figure to the currency unit. Everything about the output signals rigour — the curve, the percentiles, the tornado chart — and none of that rigour is in the inputs. **Simulation transports precision from the arithmetic to the judgement, where it does not belong.**

AI has a genuine contribution to make to the first failure and a strong tendency to worsen the second. Treating identification and quantification as separate problems, with separate controls, is the whole method of this document. The register itself is owned by `BPG-16 — Risk registers that work` and the analysis technique by `BPG-17 — Quantitative schedule risk analysis`; what follows is the AI discipline around them.

2. Identification: what machine reading genuinely adds

Candidate risks from analogous history. Where completed projects have registers and, better, records of which risks actually occurred, a model can propose the risks that materialised on similar work and are absent from this register. The output is a list of candidates for a human to accept, reject or reword — and the rejections matter as much as the acceptances, because they are where the reasoning is recorded.

Risks visible in correspondence. Requests for information (RFIs), non-conformance reports (NCRs), site instructions, minutes and letters carry the early signals of most cost and delay risk: a supplier querying a delivery date, a repeated interface query, an access restriction mentioned in passing. Machine reading covers the whole corpus rather than the sample a human can manage, and that coverage is the strongest argument for AI anywhere in risk work.

Structural gaps. Cross-reading the register against the schedule and estimate surfaces omissions mechanically: long-lead packages with no supply risk, driving-path activities with no risk attached, single-source suppliers with no alternative, consents unrepresented. These are not clever inferences; they are joins nobody has time to do by hand.

Hygiene. De-duplicating near-identical entries, merging two packages' descriptions of the same event, standardising categories and rewriting entries into a disciplined cause–event–effect form are tasks a language model does well and a busy risk manager does last.

3. What identification cannot see

Genuinely novel risk. A model proposes what resembles what it has read. First-of-a-kind work, a new contracting structure, an unprecedented regulatory change or a technology with no delivery record produce nothing to resemble.

Your own blind spots, reproduced. This is the limit that matters most and is named least. A model trained or grounded on your organisation's registers learns your organisation's habits — including the categories you never populate and the risks your culture does not write down. It will return a register that looks complete by your own historical standard, which is precisely the standard in question.

What people choose not to say. Some risks are absent because raising them is uncomfortable: a sponsor's unrealistic date, a partner's capability, a decision already taken. No document contains them, so no model can read them. Surfacing them remains a matter of trust between people, and it is one reason the workshop survives.

Cause. A model can observe that projects with a particular characteristic overran. It cannot establish that the characteristic caused the overrun, and a risk register built on association will attach mitigations to correlates rather than to causes.

4. Quantification: contribution and temptation

The legitimate contributions are real and largely mechanical: proposing three-point ranges from historical spread for review; fitting distributions and reporting the fit; running the simulation with correlations; producing sensitivity and driver rankings; tracking drawdown against the exposure remaining. All of that is computation, and computation is what machines are for.

The temptation is equally specific. Six ways a quantified output flatters itself:

Precision that outruns the inputs. A register scored by judgement in bands returns a P80 to the nearest currency unit. Reporting it that way is a claim about the analysis that the analysis cannot support. Round to the precision the inputs justify and say what that precision is.

Default ranges presented as evidence. Ranges of plus or minus ten per cent applied across a register because that is the template's default produce a distribution built on a habit. Every range should record where it came from: measured spread, supplier quotation, owner judgement, or default — and the proportion that are defaults should be reported alongside the result.

Zero correlation, unexamined. Independence is the default in most tools and is almost never true: productivity risks move together, weather affects several activities, a single supplier fails once and appears in four risks. Independence narrows the distribution and lowers the upper percentiles, which is to say it makes the answer more comfortable and less true. §8 quantifies this on a small register.

A distribution fitted to too few points. Fitting a shape to five historical observations is curve drawing, not estimation. Where evidence is thin, a simple triangular or uniform range honestly labelled as judgement is more defensible than a fitted curve that implies data.

Iteration counts that are not checked for stability. The test is not a number of iterations but whether the reported percentiles are stable when the simulation is re-run. Run it twice; if the P80 moves materially between runs, the result is noise at the precision being reported.

Risks excluded because they cannot be modelled. The unquantifiable risk — a consent refused, a partner withdrawing — drops out of the register because it does not fit the arithmetic, and the total exposure silently excludes the things most likely to matter. Carry them separately and report them alongside the number, explicitly.

To this add the failure mode AI introduces directly: a model asked to **explain** a simulation output will produce a fluent explanation whether or not one exists. A narrative that says the P80 is driven by supply chain and weather is a hypothesis to be checked against the sensitivity output, not a finding.

5. What must not be decided by a model

The confidence level. Whether contingency is set at P50, P80 or elsewhere is a statement of the organisation's risk appetite, made by the accountable authority — not a modelling parameter and not a convention to be inherited from a template.

Contingency release. Drawdown is a management decision against defined criteria, minuted, with the remaining exposure reassessed. A model may report the balance and the exposure; it may not release.

Risk acceptance. Deciding to carry a risk untreated is an accountable judgement about appetite and capacity, not an output of a score.

Whether a risk is closed. Closure means the exposure has genuinely gone — not that the date passed or that nobody has updated the entry. A model may flag stale entries; an owner closes them.

Ownership. Every risk has a named owner who can act. Assignment is a management act; a model proposing an owner from a pattern in the register is proposing, not deciding.

6. Validating a quantified risk output

1. **Confirm the register is a register.** Entries are forward-looking, in cause–event–effect form, with named owners. Issues masquerading as risks are removed to the issues log before anything is quantified.
2. **Trace the inputs.** For each of the largest contributors, confirm probability and impact with the owner, and record the evidence class: measured, quoted, judged or default.

3. **Report the default share.** State what proportion of ranges are template defaults. If it is high, the distribution is a picture of the template.
4. **Test the correlation assumption.** Re-run with a plausible alternative correlation and report the movement. An assumption whose alternative moves the answer materially is a finding, not a setting.
5. **Check stability.** Re-run and confirm the reported percentiles are stable at the precision being reported.
6. **Sanity-check against a deterministic sum.** Compare with the sum of expected monetary values (EMV). The two answer different questions, and the relationship between them should be explicable — see §8.
7. **Check for double counting.** Risk allowances embedded in the estimate, contingency held at package level, and provisions in the forecast frequently overlap. Reconcile explicitly, once, and record it.
8. **List what is excluded.** Every risk left out of the quantification, and why, reported with the result.

7. How this goes wrong

The output is adopted because it is quantified. A number with a distribution behind it beats a number without one in most meetings, regardless of which rests on better inputs. The defence is to present the evidence class of the inputs alongside the result every time.

The model's candidate risks are accepted wholesale. Two hundred proposed risks are pasted into the register to demonstrate thoroughness. The register becomes unusable, owners disengage, and real risks are buried among plausible ones. Accept candidates one at a time, with a reason.

Historical registers are treated as historical outcomes. A register records what people thought might happen, not what did. Learning from registers alone teaches a model the profession's anxieties rather than its experience. Learning from realised risk requires recording, at closeout, which risks actually occurred — a discipline most organisations lack and can start this year.

Correlation is set once and never revisited. The assumption is made during set-up by whoever built the model, is not visible in the output, and survives every subsequent review because nobody knows it is there.

The tornado is read as a cause list. A sensitivity ranking shows which inputs move the answer, and that depends on the ranges assigned as much as on the risks themselves: a risk with a wide default range will outrank a real risk with a tight measured one.

Contingency drifts into a target. Once a P80 figure is in a budget, pressure to reduce it produces quiet changes to input ranges rather than an honest argument about appetite. Version the register, and require that any change to an input range carries the owner's name and reason.

Confidentiality is lost through the register. A quantified register is a map of where a project is weakest and what the organisation believes about its suppliers. It goes into governed tools only.

8. Worked example — EMV, simulation and one unexamined assumption

Illustrative figures. A five-line extract from a quantified cost risk register. Currency USD, all figures at current price basis, impacts stated as most likely cost of occurrence. Not benchmark data.

Risk	Probability	Impact	Expected monetary value
R1 Consent decision later than programmed	0.30	900,000	0.30 × 900,000 = 270,000
R2 Rock encountered in bulk excavation	0.45	400,000	0.45 × 400,000 = 180,000
R3 Single-source supplier fails to deliver	0.15	1,200,000	0.15 × 1,200,000 = 180,000
R4 Installation productivity below assumed rate	0.60	250,000	0.60 × 250,000 = 150,000
R5 Currency movement on imported plant	0.35	300,000	0.35 × 300,000 = 105,000
Sum of EMV			270,000 + 180,000 + 180,000 + 150,000 + 105,000 = 885,000

Step 1 — what the EMV sum is and is not. USD 885,000 is the probability-weighted total. It is not a contingency: no single outcome equals it, and it says nothing about the spread. Its use is as a check on the simulation, not as an answer.

Step 2 — the simulation, as first run. With impacts ranged and risks modelled as independent, the run returns **P50 = 840,000** and **P80 = 1,340,000** (illustrative outputs from the ranges assumed for this example). At the organisation's stated appetite, contingency is recommended at P80.

Uplift over the EMV sum = 1,340,000 − 885,000 = 455,000, which is 455,000 ÷ 885,000 = 0.514 = 51.4 % above the EMV sum

The relationship should be explicable, and here it is. The EMV sum of 885,000 approximates the mean of the simulated total; the P50 of 840,000 sits below it, which is the signature of a right-skewed distribution — the low-probability, high-impact entry (R3) drags the mean up without moving the median. The P80 then sits well above both, because it reaches into that tail. A P80 below the EMV sum is also possible where one rare, very large risk dominates the register: the mean is lifted by an outcome the 80th percentile never reaches. That is a result to investigate and explain, not one to assume is an error.

Step 3 — testing the correlation assumption. R2 and R4 are both ground and productivity related: if the excavation is harder than assumed, installation productivity is likely to suffer too. The analyst re-runs with a correlation of 0.30 between them, leaving every other input untouched. The result:

P80 = 1,455,000, a movement of 1,455,000 − 1,340,000 = 115,000, or 115,000 ÷ 1,340,000 = 0.086 = 8.6 % of the original recommendation

Step 4 — read what that means. A single assumption, invisible in the output and made once during set-up, moved the contingency recommendation by USD 115,000 — more than the entire expected monetary value of R5 (USD 105,000), which had a line in the register, an owner and a discussion at the workshop. The correlation had none of those.

Step 5 — what is reported. A contingency recommendation of **USD 1,450,000** — the correlated run, rounded to the nearest 50,000 because the inputs are judgement-based and do not support finer precision — presented with: the confidence level and who set it; the correlation assumption and its effect; the share of impact ranges that are owner-judged rather than measured; and a separate statement of the two risks that could not be sensibly quantified and are therefore excluded from the figure. The recommendation is made by the named risk manager; the release of any part of it is a decision for the project director against the defined criteria.

Assumptions this answer depends on. That probabilities and impacts were confirmed with owners rather than carried forward from the last update; that the impact ranges behind the simulation are as stated and were not template defaults; that the P-values were stable across re-runs; that the 0.30 correlation is a considered judgement about ground conditions and productivity rather than a demonstration figure; and that no part of these five risks is also carried as an allowance inside the base estimate.

9. Checklist — before a quantified risk output informs a decision

1. **Register hygiene done.** Forward-looking entries, cause–event–effect wording, named owners, issues removed to the issues log.
2. **Top contributors confirmed with owners** in the current period, not inherited.
3. **Evidence class recorded** for every impact range: measured, quoted, judged or default — and the default share reported.
4. **Correlation stated and tested**, with the movement from a plausible alternative reported.
5. **Stability confirmed** by re-running and comparing the reported percentiles.
6. **EMV cross-check done**, and the relationship between the EMV sum and the chosen percentile explained.
7. **Double counting reconciled** across base estimate allowances, package contingency and the forecast.
8. **Exclusions listed** — every risk left out of the quantification and why.
9. **Precision honest** — rounded to what the inputs support, with the confidence level and its owner stated.
10. **Decisions attributed.** The confidence level, the recommendation and any release each carry a name.

Related

- `AIG-04 — AI-assisted cost forecasting` — where contingency and forecast meet, and the double-counting check at §6.7.
- `AIG-05 — AI in scheduling — and what must not be automated` — the schedule ranges that feed quantitative schedule risk analysis.
- `BPG-16 — Risk registers that work` — the register discipline this document assumes.
- `BPG-17 — Quantitative schedule risk analysis` — the analysis technique, in method detail.

- `TPL-10 — Risk register` — the instrument, including the evidence-class and correlation fields §6 relies on.

Sources and standards

- **ISO 31000:2018**, Risk management — Guidelines. In our own words: it describes risk management as an integrated, structured activity built on defined context, explicit criteria, and identification, analysis, evaluation and treatment steps, with information quality and human judgement treated as part of the process rather than as inputs to a calculation. Its insistence that criteria are set before analysis is the principle behind §5.
- **PCI Body of Knowledge, Domain 12** — Risk management for project controls (Institute manuscript, 2026). Expected monetary value, quantification and contingency setting as taught by the Institute.
- **PCI Body of Knowledge, Domain 13** — AI for project controls and project management (Institute manuscript, 2026), Knowledge Area 13.5.9, the risk workflow and its warning about contingency set by an unexamined algorithm.

Status and version

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AIG-07 · AI IN PROJECT CONTROLS GUIDE

AI in document control and correspondence

Drafting, transmitting and triaging project correspondence without conceding a position

VERSION 1.0 · DRAFT
2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

Summary

Document control is where AI meets the contract. This guide separates the correspondence AI may safely draft from the correspondence it may not, names the words and omissions that concede entitlement, sets out which register fields a model may populate and which it may never touch, and shows how to use AI to triage incoming correspondence for notification windows without letting the model become the control.

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Contents

AI in document control and correspondence

In one paragraph. Document control is where AI meets the contract. This guide separates the correspondence AI may safely draft from the correspondence it may not, names the words and omissions that concede entitlement, sets out which register fields a model may populate and which it may never touch, and shows how to use AI to triage incoming correspondence for notification windows without letting the model become the control.

Who this is for. Document controllers, contract administrators, commercial managers and project controls managers who own the correspondence register, the transmittal log and the submittal schedule — and the project managers who sign what leaves the office.

1. Correspondence is an instrument, not admin

A project's correspondence file is evidence. Two years after the event, an adjudicator, an auditor or a claims consultant reads it in date order and reconstructs what the parties knew, when they knew it and what they agreed. Every letter is a contemporaneous record that will be read by someone whose interests are opposed to yours.

So AI in document control is not the low-risk application it appears to be. The obvious framing — "it is only paperwork, the risky AI is in the forecasting" — is backwards. A wrong forecast is corrected next month. A letter saying "we accept that the delay to the pump house arose from our sequencing" is not corrected next month; it is quoted in a claim.

Two properties of assistant-class language models make this specific rather than theoretical.

First, **they draft in a cooperative register by default.** Models tuned to be helpful produce prose that concedes, softens, apologises and proposes accommodation, because that is what most correspondence in their training material does. Contract administration requires the opposite instinct: reserve, state the clause, record the fact, claim the right. The default is not neutral, and it does not read as wrong — it reads as reasonable, which is exactly how it survives review.

Second, **they complete patterns.** Asked to draft a notice, a model produces something notice-shaped: a clause number that looks right, a notification period that looks right, a recipient that looks right, because those elements belong in the pattern. Whether any of them matches your contract is a separate question the model cannot answer unless the contract is in front of it and grounded — see `AIG-03 — Data readiness: what AI needs before it is any use`.

2. Where AI genuinely earns its place

The value in document control is mostly in reading, not writing. Four applications are worth the governance they cost.

Incoming triage at coverage. A model reads every incoming item — letters, site instructions, requests for information (RFIs), minutes, emails — and flags those that may start a clock, create an obligation or signal a variation. Its value is coverage: it reads everything, every day, without fatigue. Its output is a queue for a human, never a decision. See §6 for the arithmetic of how good a triage screen has to be.

Register enrichment. Populating descriptive metadata on a document register — subject, discipline, originating party, referenced drawing numbers, referenced clauses, related transmittal — from the document itself. This is extraction, the capability class AI is strongest at.

Retrieval across the correspondence set. "What did we write to the subcontractor about access to level 3, and when?" answered from the actual correspondence, each answer cited to a document reference. Ungrounded, this question invites a fabricated letter; grounded in the document set with citations opened and checked, it replaces a day of searching — and it summarises a forty-item RFI chain into a chronology a project manager can read before a meeting.

Submittal and transmittal status reasoning. Comparing the submittal schedule to the transmittal log and flagging items that are overdue, superseded, or approved-with-comments and never resubmitted — the least glamorous and most reliably useful item on the list.

Everything here is proposal. None of it is release.

3. Four classes of outgoing correspondence

Not all outgoing document is equally exposed. Classify it before you decide what AI may touch. The class determines the drafting rule, not the author's confidence.

Class	Examples	May AI draft?	Release rule
A — Transmittal and routine	Drawing transmittals, document issue sheets, distribution notes, acknowledgements of receipt with no substantive content	Yes, from the register data	Document controller checks the register fields against source; no free text beyond the standard form
B — Technical and informational	Covering letters to submittals, progress narratives to the client, meeting minutes, responses to technical queries with no commercial content	Yes, as a first draft	Discipline lead verifies technical content; controls manager checks no commercial or entitlement statement has entered the text
C — Commercial	Valuation covering letters, variation quotations, responses to a payment assessment, correspondence about rates or quantities	Draft only from a controlled template and only where the numbers come from a verified source	Commercial manager verifies every figure to source and signs; the AI's role is limited to assembling the controlled wording
D — Notices and contractual positions	Notices of delay, notices of a compensable event, claims correspondence, responses alleging breach, anything relying on a clause or reserving a right	**No** — not drafted by a model	Drafted by the person accountable for the position, on the contract's own wording, with legal review where the exposure warrants it

The C/D boundary is where most functions get into difficulty, because a valuation covering letter can become a contractual position in a single sentence. The practical rule: **if the letter relies on a clause, asserts or resists an entitlement, characterises a cause, or agrees anything, it is class D**, whatever the register calls it.

Class D is not a prohibition on AI in the vicinity. A model may retrieve the relevant clauses for the drafter, assemble the chronology from the register, and check a human-drafted notice against a compliance checklist. It may not produce the words that go out.

4. The sentences that concede

An AI-drafted letter concedes in ways that are easy to read past because they are grammatical, polite and plausible. These are the recurring patterns worth training reviewers to catch.

Admissions of causation. "Following the delay caused by our late issue of the reinforcement drawings..." The model wrote a causal link because the surrounding text implied one. Causation is the contested issue in almost every delay dispute; it is settled by analysis, not by a subordinate clause in a covering letter.

Unqualified acceptance. "We accept the revised programme." Accepting a programme can accept its logic, durations and sequencing constraints, and can undermine a later position that the programme was unachievable. The reserved form — "we acknowledge receipt; our comments follow and our rights under clause [x] are reserved" — is a different letter.

Apology as admission. "We apologise for the delay in returning the submittal." Where the return period is contractual and the delay is disputed, this sentence is a gift.

Waiver by generosity. "In the interests of collaboration we will absorb the additional cost on this occasion." A model asked for a "constructive tone" produces this unprompted.

Characterisation that defeats a claim. "This is a minor adjustment and we do not expect it to affect the programme." Said about work that later needs an extension of time (EOT), this is the sentence the other side reads out first.

Blanket no-cost-no-time confirmations. "We confirm there is no cost or time implication." Correct only where it has been assessed; a model produces it because it is a common closing formula.

Defective notices. Most notice regimes require a specific set of elements: the clause relied on, the event, the date the party became aware, the effect claimed, and delivery by the contractual method to the named recipient within a stated period. An AI-drafted notice typically reads well and satisfies three of the five. A notice that reads beautifully and goes to the wrong address is void.

Silent scope expansion. A drafted RFI response that answers the question and then helpfully explains what the contractor "should" also do has instructed work.

The reviewer's discipline is not "read for tone". It is: **read every verb whose subject is your own organisation, and every sentence containing a causal connective, and ask whether you would sign it standing in front of an adjudicator.**

5. The register: which fields a model may populate

A document register's fields are not equal. Split them, and put the split in the procedure.

Model may populate, human samples: subject line, discipline, document type, originating party, referenced drawing and specification numbers, referenced correspondence, keywords, summary text, proposed distribution list.

Model may propose, human confirms every instance: the reply-required flag, the response due date, the contractual clause referenced, the link to a variation or claim reference, the risk or issue register link.

Human only, never model-populated: the date of receipt of record, the notice classification, the notification window expiry, the transmittal number, the revision status, the approval status, and any field a downstream calculation or a legal deadline reads.

The rule behind the split: **a model may populate a field a human will read; it may not populate a field a system will act on.** A wrong keyword costs a search. A wrong notification-window expiry costs the claim.

6. Incoming triage: recall matters more than precision

For an incoming-correspondence screen, the two error rates are not symmetric. Precision asks how many of the flagged items really mattered — the cost of false alarms is reviewer time. Recall asks how many of the items that really mattered were flagged — the cost of a miss is a lapsed notification window.

A screen that flags too much wastes hours. A screen that misses one letter loses an entitlement. Tune, evaluate and accept a triage model on **recall**, and treat its precision purely as a workload number.

The design consequence functions get wrong: the triage model is not the control. The control remains the contractual notice diary — the independently maintained list of every obligation with a date, owned by the contract administrator and reconciled to the register at a stated frequency. The model shortens the reading; the diary catches the miss. Where a model is the only line of defence, the function has replaced a control with a convenience.

7. Confidentiality across the document set

A project document set is among the most commercially sensitive data an organisation holds: rates, subcontract terms, claims strategy, personal data in correspondence, and in some sectors security information. Two rules follow.

Nothing leaves the governed environment. Contracts, commercial correspondence and claims material are processed only in tools approved for that data class in the permitted-use register — see `AIG-08 — Governing AI on a project — the control framework` §3.

Retrieval must respect the permission model. A retrieval system indexed over the whole document management system will answer from material the asker is not entitled to see — a claims strategy note surfaced to a joint-venture partner's user, a personnel matter surfaced to a site engineer. Before it is switched on across a project, the test is not "does it answer well" but "does an unprivileged account get an unprivileged answer". Run that test with a named unprivileged account and record the result.

8. How this goes wrong

The letter nobody read as a legal document. A class C letter is drafted by a model, reviewed by someone checking the numbers, and signed by someone checking the letterhead. Nobody read it as a contractual instrument. The concession is found eighteen months later by the other side's consultant.

The clause number that came from nowhere. A drafted notice cites a clause that is a plausible place for a notice provision in a common standard form, but not the clause in this contract. The reviewer, who expected the same clause, did not open the contract.

The triage model that became the control. Six months of accurate flagging builds confidence, the notice diary quietly stops being reconciled, and the one letter the model does not recognise — a scanned attachment with unusual phrasing — is never flagged. The failure is invisible until the window has closed.

Register poisoning at scale. A bulk enrichment run populates 4,000 register entries with model-derived metadata. Two per cent are wrong, nobody knows which two per cent, and the register is what downstream reporting draws on. Bulk runs need a sampled acceptance test before the write.

Retrieval that answers from the wrong version. A grounded model answers a specification question from revision B while revision D governs, because the index holds both and nothing said which is current. Every retrieval answer in document control carries a document reference and a revision, and the reviewer checks the revision rather than merely that a citation exists.

Minutes that create agreements. A meeting assistant drafts minutes recording that a party "agreed" to something they discussed. Circulated unchallenged, minutes become evidence of agreement. Minutes are class B correspondence with a class D tail: actions and any agreement language are confirmed with the named owners before circulation.

Volume mistaken for diligence. AI makes it cheap to write more letters. More letters is not a better correspondence position; it is a longer file for the other side to mine.

9. Worked example — sizing a triage screen

Illustrative figures. Figures are per calendar month for a single project; time is reviewer minutes; rounding to the nearest whole item.

Setup. The project receives **480** incoming correspondence items a month. Of these, **45** are genuinely entitlement-bearing — they start a clock, create an obligation or signal a variation. A triage model flags **62** items, of which **41** are genuinely entitlement-bearing. Reviewing one item takes **6 minutes**.

Formulae. `precision = true flags ÷ total flags`; `recall = true flags ÷ true cases`; `review minutes = items reviewed × minutes per item`.

Substitution.

- Precision `= 41 ÷ 62 = 0.661 = 66.1 %`
- Recall `= 41 ÷ 45 = 0.911 = 91.1 %`
- Missed items `= 45 − 41 = 4`
- Unflagged items `= 480 − 62 = 418`; a 10 % residual sample `= 41.8 → 42` items
- Review effort with the screen `= (62 × 6) + (42 × 6) = 372 + 252 = 624` minutes `= 10.4` hours
- Review effort reading everything `= 480 × 6 = 2,880` minutes `= 48.0` hours

Result. The screen cuts the monthly reading from 48.0 hours to 10.4 hours, at a recall of 91.1 % — and leaves **4 entitlement-bearing items unflagged every month**.

Interpretation. The saving is real and the residual exposure is unacceptable on its own terms. The 10 % sample is expected to catch `4 × 10 % = 0.4` of the four misses — it will usually catch none of them. The sample is a drift detector, telling you whether recall is degrading; it is not a safety net. The safety net is the contractual notice diary of §6, reconciled independently. The case for the screen is workload; the case against relying on it is the four items — both true at once, and both stated to whoever approves the deployment.

10. Checklist

Take this into the meeting where AI in document control is proposed.

- [ ] Every outgoing document type is classified A–D (§3), and the classification is in the procedure, not in someone's head.
- [ ] Class D is drafted by the accountable person, not by a model, and the procedure says so.
- [ ] The reviewer of any AI-drafted letter has been briefed on the concession patterns in §4.
- [ ] Register fields are split into model-populated, model-proposed and human-only (§5), and the human-only list includes every field a calculation or a deadline reads.
- [ ] The triage model has a measured recall on a dated sample, and the number is known to whoever approved it.
- [ ] The contractual notice diary exists, is owned by a named person, and is reconciled to the register at a stated frequency.
- [ ] Retrieval has been tested with an unprivileged account and the result recorded.
- [ ] Every retrieval answer carries a document reference **and** a revision.
- [ ] Bulk enrichment runs have a sampled acceptance test before the write.
- [ ] Meeting minutes' actions and any agreement language are confirmed with named owners before circulation.
- [ ] Contracts and commercial correspondence are processed only in tools approved for that data class.

The correspondence file will be read by someone who is not on your side. Write it for that reader, and decide in advance which sentences a machine is allowed to contribute.

Related

- `AIG-08 — Governing AI on a project — the control framework` — the permitted-use register, data classes and verification tiers this document assumes are in place
- `AIG-10 — Human in the loop: what AI may and may not decide` — the decision boundary that puts contractual positions on the human side, and the authority thresholds that go with it
- `AIG-12 — The AI-literate controls professional` — how a reviewer builds the instinct to catch a conceding sentence
- `BPG-11 — Change orders and variations` — the variation process the correspondence register feeds
- `BPG-12 — Claims and extension of time` — what the correspondence file has to survive

Sources and standards

- PCI Body of Knowledge, Domain 7 (Contracts, Commercial Management, BoQ, Invoicing & Revenue) and Domain 13 (AI for Project Controls & Project Management), `docs/bok/` — the source material this series draws on; explained here in our own words, not reproduced.
- The Institute's candidate AI-use policy (`docs/downloads/`) — the governing position on AI in preparation, examination and practice.

No external standard is reproduced in this document. Notice provisions, notification periods and delivery requirements differ between contract forms and jurisdictions: this document describes the discipline, and your contract states the rule.

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AIG-08 · AI IN PROJECT CONTROLS GUIDE

Governing AI on a project — the control framework

Permitted uses, approval authority, records, confidentiality and model change control, written as project controls

VERSION 1.0 · DRAFT

2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

Summary

A corporate AI policy does not govern a project; project controls do. This document sets out the six control objects that make AI use on a project defensible — permitted-use register, verification tiers, approval authority, verification record, data classification and tool-and-model change control — gives the clauses to write into the project execution plan, and shows how to size and budget the verification effort the framework creates.

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Who this is for. Project controls managers, project directors, PMO leads and heads of function who answer for AI-assisted numbers, and the assurance and audit staff who test the answer.

1. The gap a corporate policy leaves

Most organisations that have adopted AI have written a policy: confidential data must not go into public tools, outputs must be verified, staff remain accountable. All correct, and none of it governs a project.

A project needs to know: on this project, which task may use AI, on which data, in which tool, verified to what standard, approved by whom, recorded how, retained how long. A corporate policy answers none of those questions. It states a principle; a project needs a control.

It is the distinction controls professionals already make about cost. "We will manage cost carefully" is a policy; a cost breakdown structure, a change control procedure, delegated authority thresholds and a monthly reconciliation are controls. AI governance fails for the reason cost control fails: the principle was written and the control was not.

This document is the control set — the Institute's position, **AI proposes; the professional disposes**, turned into artefacts with owners, frequencies and fields.

2. Six control objects

Everything that follows attaches to one of six objects. A framework with all six is auditable; one with fewer should name which is missing, and why.

#	Control object	What it fixes	Owner (typical)	Cadence
1	**Permitted-use register**	Which task, capability class, data class and tool	Project controls manager	Reviewed at each stage gate; entries added on request
2	**Verification standard (tiers)**	How hard an output is checked before use	Project controls manager	Set at mobilisation; changed by documented decision only
3	**Approval authority**	Who may decide what, at what threshold	Project director	Set at mobilisation; aligned to delegated authority
4	**Verification record**	Evidence that the check happened	Output owner (named)	Per material output
5	**Data classification**	What may go into which tool	Information manager	Inherited; mapped to project artefacts at mobilisation
6	**Tool and model change control**	What happens when the tool changes under you	Named tool owner	Re-validation on trigger and on a stated calendar

Objects 3 and 4 are owned elsewhere in this series: the decision boundary and authority thresholds in `AIG-10 — Human in the loop: what AI may and may not decide`, the audit-trail content in `AIG-09 — Bias, explainability and auditability`. This document owns objects 1, 2, 5 and 6, and the assembly of all six into the project execution plan.

3. The permitted-use register

The register is the operative document — not a list of approved products but of **approved uses**, because the same tool is appropriate for one task and reckless for another. One row per use:

Column	Content	Why it is there
Use reference	`AI-U-nn`	So a record or an audit finding can cite it
Task	The controls task in the function's own words, e.g. "draft first-cut control account variance narrative"	A task, not a capability: "summarisation" is not a use
Capability class	Assistant, retrieval-grounded, tabular analysis, forecasting model, process automation, transcription	Ties the row to its class-specific failure mode (`AIG-02`)
Approved tool	The named instance, and whether it is the governed enterprise deployment	The same product ungoverned is a different control
Maximum data classification	The highest classification permitted in this use	The most commonly breached rule
Grounding required	Yes/no, and against what corpus	Grounded and ungrounded answers differ in risk
Verification tier	1, 2 or 3 (§4)	Fixed in advance, not chosen under deadline
Output owner role	The role accountable for this use's outputs	Accountability precedes the output
Approved on / review due	Dates	Enforces §6 re-validation
Status	Approved · Provisional (pilot) · Suspended	A suspended row beats a deleted one

Three rules give the register its force. **Absence is prohibition** — a use not in the register is not permitted, the only formulation that survives contact with a busy team, because "use your judgement" produces an unbounded surface with no audit trail. **Provisional means measured** — a pilot enters as Provisional with a success measure and an end date, and a pilot that misses its measure is closed, not extended (`AIG-11` §8). **Requests are cheap and logged** — a framework people cannot add to is one they work around, so provide a request route to the register owner and log refusals with reasons. The refusal log is the most informative document in the framework after six months.

4. Verification tiers

The commonest governance failure is a uniform rule — "all AI outputs must be verified" — which is either unaffordable or ignored. Set three tiers; assign one to every register row.

Tier 1 — Independent recomputation. The reviewer reproduces the number by an independent route without reference to the model's working, then compares; assumptions, method and the reason for the method are recorded. Applied to anything reported outside the project, anything entering a baseline or a decision-funding forecast, and anything with contractual effect.

Tier 2 — Source verification. Every figure and extracted item traced to its source record, every causal claim checked against the underlying analysis, and anything the model could not ground deleted rather than softened. Applied to internal analysis, extraction, narrative drafting over verified data and register enrichment.

Tier 3 — Sampled acceptance with monitoring. High-volume, low-unit-consequence output — coded transaction lines, metadata enrichment — accepted on a defined sample with an error threshold and monitored for drift. Sample size, threshold and the action on breach are written down before the first run.

Three points decide whether tiers work.

- **The tier belongs to the use, not the day.** It is set in the register when the use is approved. Nobody chooses their own tier at 17:00 on a reporting day.
- **A tier is lowered only by documented decision** from the register owner, on measured error rate — not on time elapsed without incident. It is a control change, logged as one.
- **Tier 3 is not "no verification".** It swaps per-item review for a sample plus a monitoring obligation, and the monitoring is what gets dropped. Name the person who reads the drift number, and the month.

5. Confidentiality of commercially sensitive data

Four things make project confidentiality more than a compliance formality: unit rates and build-ups, subcontract terms and commercial positions, claims and dispute strategy, and personal data in correspondence and resource records. In joint ventures, some is confidential between the parties on your own project.

Map the classification before mobilisation, not at first use. Write down which project artefacts fall into which class of the organisation's scheme: cost model, risk register, subcontract set, correspondence file, resource plan, claims file. The register's "maximum data classification" column is meaningless until this map exists.

The pre-entry test. Before material enters any tool, the person entering it answers three questions: what is this material's classification; is this tool approved for that classification; would the data owner be content to watch me do this. The third catches what the first two miss — notably joint-venture and client data whose classification nobody has set.

Preparation is a control. Removing names, rates or party identifiers before analysis often converts a prohibited use into a permitted one at negligible cost; record what was removed, because the analysis's limitations follow from it. A retrieval system indexed across the project must inherit the underlying access model, tested with a named unprivileged account before go-live. Where data is processed, under whose law, and whether it improves the vendor's models are settled at purchase (`AIG-11` §3): the register records those answers, it does not discover them.

6. Tool and model change control

This is the control object most frameworks omit, and the one that ages an otherwise sound framework into a false assurance.

The system you validated is not the system you are using. Hosted models are updated by their vendors, platform features change on the vendor's release cycle, a retrieval corpus changes every time someone files a document, and a forecasting model's world changes as the portfolio changes. Nothing in your project changed, and the behaviour did. Four controls follow.

Version identification. The verification record captures the tool, and the model or version identifier where the vendor exposes one. Where it exposes nothing — common for embedded platform features — record that: "version unknown" is a finding, and caps how far a number can later be reconstructed.

Re-validation triggers. Re-validate on any of: a vendor-notified model or feature change; a change to the retrieval corpus's scope or permissions; a change to an upstream data structure such as the cost breakdown structure or code of accounts; a project stage gate; a failed sample under Tier 3; and a stated calendar cadence for anything at Tier 1. Write the trigger list into the plan.

Re-validation method. Re-run a versioned set of test cases whose correct answers were established by professionals and compare with the previous run — the same instrument used to evaluate the tool at purchase, kept and re-used. `AIG-09` §7 covers what a change in the results means.

Change is announced, not discovered. The register's tool owner tracks vendor release notes and tells users when behaviour has changed. A user who discovers a change by getting a different answer has already relied on the old one.

7. What goes in the project execution plan

The framework lives in the project execution plan, or the controls plan where the function keeps one. Ten clauses, numbered so an assurance finding can cite them — see `TPL-01 — Project controls execution plan`.

AI-1 Scope. Which parties the section binds — staff, contractors, secondees, consultants and, where the contract allows, subcontractors producing controls deliverables — and that accountability for every output rests with a named individual, not a tool.

AI-2 Permitted-use register. Where it lives, who owns it, how a use is added, and that a use not in the register is not permitted.

AI-3 Data classification and prohibited data. The classification map (§5), the maximum classification per approved tool, and material that may not enter any tool on this project.

AI-4 Verification tiers. The three tiers, what each requires, and that a tier is assigned in the register and lowered only by documented decision on measured evidence.

AI-5 Approval authority and sign-off. The decision boundary and thresholds, cross-referenced to the project's delegated authority schedule so the two cannot diverge. See `AIG-10`.

AI-6 Disclosure. Where AI assistance is disclosed — client, board, auditor, certifier — and in what form. Contract and reporting requirements govern; where they are silent, the project states its own rule rather than leaving it to individuals.

AI-7 Records and retention. What a verification record contains, where it is filed, and for how long. Retention matches the project's records schedule: a number that outlives its evidence is undefendable.

AI-8 Tool and model change control. The four controls of §6, with the named tool owner and the trigger list.

AI-9 Incidents and near-misses. What is reported, to whom, within what period; that a near-miss is reported as readily as an incident; and that the register and plan are updated where one shows a gap.

AI-10 Review cadence and ownership. Who owns this section, when it is reviewed, and that everyone using AI on the project is briefed before first use.

8. Knowing whether the framework is working

Three tests, none of which is "we have a policy". **Reconstruction:** pick an AI-assisted number that left the project last quarter and ask its named owner to reconstruct it — source data, method, what the model produced, what changed in review, who signed, when; more than an hour means the record is not working. **Refusals:** no refusals logged in six months means the framework is unused or unconsulted, both findings. **Incidents:** zero incidents and zero near-misses across an active function means near-misses are not reported, and the first thing you learn about will be an incident that reached the client.

9. How this goes wrong

Policy without register. An AI policy exists and no project-level permitted-use register. Everyone is "being careful", nobody can say which tasks use AI on what data, and nothing is auditable or prohibited.

The register nobody can add to. Additions require a monthly committee. Within one reporting cycle staff have gone around it, and the shadow use is unmonitored and invisible — worse than the use the framework meant to prevent.

Uniform verification, quietly abandoned. "Everything is verified" survives four weeks. Because no tier distinguishes a board forecast from a metadata field, the rule is impossible, so it is ignored — and it is ignored for the board forecast too.

Verification recorded as a tick. "Verified — yes" proves nothing; under challenge the reviewer cannot say what they recomputed or against what. `AIG-09` §3 gives the fields that make a record load-bearing.

Governance costed at zero. The business case counted licences and time saved, not verification hours, so the framework arrives with no budget and is treated as optional. §10 prices this deliberately.

The silent upgrade. A tool's model is updated, outputs shift, and nobody re-validated because nothing in the project changed. Three months of numbers rest on an assurance that expired without notice.

Confidentiality by exhortation. The rule is "no confidential data in public tools", but the project never classified its artefacts, so staff judge sincerely about material they have no classification for — and the joint-venture partner's rates end up in a public assistant.

The framework as a shield. Documents exist, boxes are ticked, and nobody has refused an output, re-validated a tool or investigated a near-miss. Worse than nothing: it manufactures confidence.

10. Worked example — pricing the verification the framework creates

Illustrative figures. One project, one month, then annualised. Currency stated as USD; loaded labour rate assumed; rounding to the nearest whole unit.

Setup. The register lists nine uses producing material outputs monthly: **four at Tier 1** (cost forecast, contingency drawdown analysis, schedule-risk input pack, client cost report) and **five at Tier 2** (variance narratives, register enrichment, contract-term extraction, exception commentary, risk-register update). Tier 1 verification averages **90 minutes** per output, Tier 2 **30 minutes**. Loaded cost of the reviewing grade: **USD 85 per hour**.

Formulae. `monthly minutes = (Tier 1 count × Tier 1 minutes) + (Tier 2 count × Tier 2 minutes)`; `annual hours = (monthly minutes ÷ 60) × 12`; `annual cost = annual hours × loaded rate`.

Substitution. Tier 1 `= 4 × 90 = 360` minutes; Tier 2 `= 5 × 30 = 150` minutes; monthly `= 360 + 150 = 510` minutes `= 510 ÷ 60 = 8.5` hours; annual `= 8.5 × 12 = 102` hours; annual cost `= 102 × 85 = USD 8,670`.

Result. The framework costs **102 reviewer-hours and USD 8,670 a year** on this project, before any tool licence.

Interpretation. The number does three jobs. It goes in the controls budget as a line, so verification is resourced rather than expected. It goes in the value case as a cost netted against whatever the AI use saves — a case that omits it is not a case. And it is the honest test of scope: at thirty material outputs a month, verification is `(4 × 90) + (26 × 30) = 360 + 780 = 1,140` minutes, or **19 hours a month**, at which point the function resources it, moves lower-consequence uses to Tier 3 with monitoring, or removes uses from the register. What it must not do is keep the register and abandon the tier — §9's third failure.

11. Checklist

- [ ] A permitted-use register exists for this project, with an owner and a review date.
- [ ] Every row states task, capability class, tool instance, maximum data class, grounding, tier and output-owner role.
- [ ] "A use not in the register is not permitted" is written in the plan; a request route exists; refusals are logged.
- [ ] Three tiers are defined, every row carries one, and lowering a tier needs a documented decision on measured evidence.
- [ ] The data classification map names actual project artefacts, and each tool has a maximum classification plus recorded residency and training-on-your-data answers.
- [ ] Retrieval permissions have been tested with a named unprivileged account.
- [ ] Each tool has a named owner, a re-validation trigger list and a versioned test set.
- [ ] Verification records are filed where an auditor can find them, retained to the project's records schedule.
- [ ] Clauses AI-1 to AI-10 are in the plan, and verification effort is a costed line in the controls budget.
- [ ] An incident and near-miss route exists, and someone has read the log this quarter.

When an AI-assisted number is challenged — by a client, an auditor, an adjudicator or a board — the answer should be a record and a name produced within the hour, not a search for who ran what, in which tool, on which day.

Related

- `AIG-09 — Bias, explainability and auditability` — what a verification record must contain to make a number reconstructable
- `AIG-10 — Human in the loop: what AI may and may not decide` — the boundary and thresholds control object 3 depends on
- `AIG-11 — Evaluating AI tools — a buyer's due-diligence guide` — how residency, training and versioning answers are obtained
- `ETH-05 — The ethical use of AI and data` — the conduct obligations underneath the controls
- `TPL-01 — Project controls execution plan` — the plan these clauses go into

Sources and standards

- PCI Body of Knowledge, Domain 13 (AI for Project Controls & Project Management), `docs/bok/` — the governance and assurance material this document draws on, explained in our own words.
- The Institute's candidate AI-use policy (`docs/downloads/`).

Standards named but not reproduced: ISO/IEC 42001 (management systems for artificial intelligence) and ISO 31000 (risk management), the natural reading for a function formalising this at organisational rather than project level. This is a project controls framework, not a management-system standard, and not a route to conformity with either.

Status and version

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AIG-09 · AI IN PROJECT CONTROLS GUIDE

Bias, explainability and auditability

If you cannot reconstruct how a number was produced, you cannot defend it

VERSION 1.0 · DRAFT
2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

Summary

Bias, explainability and auditability are controls problems, not ethics-seminar topics. This document sets the standard a project must meet — an AI-influenced number that cannot be reconstructed cannot be defended in a claim or an audit — then gives the fields of a reconstruction record, the places bias actually enters a controls workflow, how to test for skew and drift, and what to do when a tool cannot be explained at all.

About this document

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Contents

Bias, explainability and auditability

In one paragraph. Bias, explainability and auditability are controls problems, not ethics-seminar topics. This document sets the standard a project must meet — an AI-influenced number that cannot be reconstructed cannot be defended in a claim or an audit — then gives the fields of a reconstruction record, the places bias actually enters a controls workflow, how to test for skew and drift, and what to do when a tool cannot be explained at all.

Who this is for. Cost engineers, planners, forecasting and risk analysts who produce AI-assisted numbers, and the controls managers, auditors and assurance reviewers who have to stand behind them.

1. The question that settles it

An AI-assisted estimate at completion (EAC) is challenged eleven months later, in a claim, an internal audit or a board paper. One question decides whether it survives: **can a competent person, using the record you kept, reproduce how that number was arrived at — and say why it is what it is rather than something else?**

That question is not about ethics or technology. It is the same evidential standard controls has always applied to a manual forecast: source data, method, assumptions, the reason for the method, the reviewer, the date. AI changes nothing about the standard. It changes how easy it is to fail it, because the working that used to sit in a spreadsheet now sits inside a system that does not keep it and cannot be re-run to the same answer.

Three things follow, and they are the substance of this document. **Explainability** is the property that makes reconstruction possible. **Auditability** is the record that makes reconstruction possible later, when memories have gone and the model has changed. **Bias** is the systematic error that reconstruction is most likely to expose, and least likely to expose by accident.

2. Three kinds of explanation, and which one is worth having

"The tool is explainable" is claimed at three quite different depths. Know which one you have.

Mechanism. An account of how the model type works — a regression on historical outturn, a classification against learned patterns, a language model predicting text. Useful for choosing a capability class; useless in a challenge. It explains the machinery, not the number.

Drivers. An account of which inputs moved this particular output, and by how much: "the forecast rose because the labour productivity input fell and the remaining duration extended". This is what most vendors mean by explainability, and it is genuinely valuable — it lets a professional judge whether the model is responding to the things that actually changed.

Reconstruction. The number can be rebuilt from stated inputs by a stated method, arriving at the same answer within a stated tolerance, without the model. This is the only level that meets the evidential standard of §1.

Reconstruction does not require the model to be simple. It requires the reported number to be attributable. In practice that means one of three arrangements: the model's method is a known formula applied to identifiable inputs, so it can be recomputed by hand; or the model's output is used as an input to a human method that is itself reconstructable; or the model's proposal is checked against a conventional calculation, and the conventional calculation is what is reported, with the model's proposal recorded as corroboration. The third arrangement is the honest answer for opaque tools, and it is treated in §8.

3. The reconstruction record

A verification record that says "verified: yes" is worthless in a challenge. These are the fields that make one load-bearing. One record per material AI-assisted output, filed with the output.

Field	Content	Why a challenge needs it
Output reference	The number or artefact, and where it was reported	Ties the record to what left the project
Permitted-use reference	The `AI-U-nn` row it was produced under	Shows the use was approved, and under which rules
Tool and version	Tool instance and model or version identifier; "not exposed" where the vendor gives none	Behaviour changes with version; without it, later re-runs prove nothing
Input data	The datasets and cut-off date or period; where they were extracted from	The commonest cause of an unreproducible number is an unrecorded data cut
Method and parameters	The method the model applied, and any parameter that changes the answer	"Which method" is the first question an auditor asks about a forecast
Assumptions	Every assumption the answer depends on, stated in the same breath as the answer	An unstated assumption is how a defensible number becomes an indefensible one
What the model produced	The unedited output, retained	Distinguishes the model's proposal from the human's judgement
What changed in review, and why	The delta and its reason	This is the field that proves a human actually reviewed it
Verification performed	Which tier (`AIG-08` §4), what was recomputed, against what source	A named check, not a tick
Named sign-off and date	The individual, not the team	A model cannot be accountable; a person can

Two rules about the record. It is written **at the time**, because a reconstruction record assembled after a challenge is worth very little and looks worse. And it is retained to the project's records-retention schedule: a number whose evidence has been deleted is, for practical purposes, unsupported.

4. What breaks an audit trail

Four failures account for most unreconstructable numbers, and all four are avoidable at negligible cost.

The moving data cut. The record names a dataset but not the period or extract date. The dataset has since been updated, so re-running the analysis gives a different answer and the original cannot be recovered. Record the extract date and, for anything at Tier 1, retain the extract itself.

The unversioned tool. The tool has been upgraded; the vendor exposes no version identifier; the record notes only the tool's name. Nobody can now say whether a different answer today reflects a data change or a model change. Where a vendor exposes nothing, record that fact — and treat "version not exposed" as a factor in the tool's suitability for Tier 1 work (`AIG-11` §3).

The lost prompt or configuration. For assistant-class and retrieval tools, the instruction given is part of the method. A record that keeps the output but not the instruction has kept the answer and thrown away the question.

The invisible human edit. The reported number differs from the model's output and nothing records the difference or its reason. Under challenge, this is the worst of both worlds: the professional cannot show they exercised judgement, and cannot show they did not simply overwrite an inconvenient result.

5. Where bias actually enters a controls workflow

Bias in a controls context is not primarily a question of protected characteristics — though where AI influences decisions about people, that question is live and is governed by employment and data-protection law that varies by jurisdiction. The dominant risk in cost, schedule and risk work is **systematic error inherited from the data**, and it has specific, findable homes.

Historical outturn that encodes one era. A model trained on projects delivered in a period of stable prices will under-forecast in a period of escalation. It is not wrong about the past; it is being asked about a future the past does not contain.

A portfolio mix that is not your project. Training data dominated by one project type, size, region or contracting model will systematically misjudge a project outside that mix. The failure is quiet: the model produces a confident number, and its confidence carries no information about how unlike its training data your project is.

Optimism embedded in the labels. If the historical record was itself produced under pressure to report favourably — approved forecasts that were consistently late to recognise overrun — a model learns to reproduce that optimism. It is a faithful model of a biased practice.

Survivorship in the record. Cancelled scopes, abandoned packages and terminated subcontracts often leave the dataset. A model trained on what completed will understate the cost of what does not.

Scoring that recycles a reputation. Subcontractor or supplier risk scoring trained on historical performance data will penalise parties who were previously given the difficult work, and reward those who were not. The score is a measure of past allocation as much as past capability, and where it influences selection it is also a commercial and legal exposure.

Assumption inheritance. Productivity rates, waste allowances and duration norms carried from prior projects into a model's inputs are assumptions, not facts, and they are rarely re-examined once they are inside a tool.

6. Testing for skew

Bias is found by segmenting, not by asking whether a tool is fair. Take the evaluation set used to accept the tool — a sample of cases whose correct answers were established by professionals and kept under version control — and score performance **by segment** rather than in aggregate.

Segment on the dimensions your work actually varies along: project type, contract form, value band, region, delivery stage, and discipline. An aggregate accuracy figure conceals exactly the failure you care about, which is that the tool performs well on the majority case and badly on the minority case — and the minority case is often the project in front of you.

Two disciplines make this real. The evaluation set is **never used to tune the model it tests**, or the score stops measuring anything. And segment results are recorded with the acceptance decision, so a reviewer six months later can see the tool was accepted knowing it was weaker on, say, refurbishment work — which is a legitimate decision, and an illegitimate surprise.

7. Drift

A model that was right last year can be silently wrong this year. Nothing announces it. The portfolio mix changes, the cost breakdown structure is restructured, a market moves, the vendor updates the model, and performance degrades while the interface looks identical.

The control is a **re-run on a cadence**: the same versioned evaluation set, scored again, compared with the previous run, on the schedule and triggers set in the permitted-use register (`AIG-08` §6). A change in score is investigated before anyone re-tunes a threshold — because re-tuning to restore the old score without knowing why it moved converts a detectable problem into a hidden one.

Drift monitoring has an owner and a date, or it does not exist. The most common form of this failure is not a missing procedure; it is a procedure with no name against it.

8. When explainability is unobtainable

Some tools will not give you reconstruction, and no amount of governance will extract it. This is a decision point, not a reason to give up on the tool.

Do not use it as the reported number. An opaque output does not go into a baseline, a board forecast, a contractual position or a disclosure — see `AIG-10 — Human in the loop: what AI may and may not decide`.

Use it as a screen, a challenge or a corroboration. An unexplainable model that flags projects for attention, or that disagrees with your conventional forecast and prompts you to look again, is useful and carries no evidential burden. What is reported is the conventional calculation; the model's disagreement is recorded as a reason the calculation was re-examined.

Say so in the record. "Model output not reconstructable; used as corroboration only; reported figure derived by [method]" is a complete and defensible entry. What is not defensible is an opaque number reported as though it were reconstructable.

9. How this goes wrong

Explainability accepted at the wrong depth. The vendor demonstrated driver attribution; the project assumed that meant reconstruction. The first time a number is challenged, the team can say which inputs moved and cannot rebuild the figure.

The record written after the challenge. Nobody kept a reconstruction record. When the number is queried, one is assembled from memory and calendar entries. It is probably accurate. It reads as constructed, because it was.

Bias discussed, never tested. The function has a thoughtful position on AI bias and has never segmented an evaluation set. Aggregate accuracy is good; performance on the two project types that make up this year's growth has never been measured separately.

Drift assumed away. The tool was validated at deployment and never re-scored, because nothing in the project changed. The vendor's model was updated twice in the period.

Sign-off by team. The record names "Project Controls" as reviewer. Under challenge, no individual can say what they checked. A model cannot be accountable and neither can a department.

Reconstruction defeated by the data cut. Every other field is complete, but the record names a live dataset with no extract date. Re-running produces a different answer, and the original working cannot be recovered.

The human edit that erased the evidence. The professional corrected the model's output — correctly — and reported the corrected figure without recording the change. The correction is now invisible, and so is the review that produced it.

Explainability treated as an ethics matter. The topic is delegated to a policy team, discussed as principle, and never turned into fields on a form. Nothing changes in how numbers are produced.

10. Worked example — reconstructing an AI-produced forecast

Illustrative figures. Currency USD; figures as at the month-end data cut; indices shown to four decimal places; the division is performed on the unrounded index and the result rounded to the nearest whole currency unit.

Setup. A forecasting tool reports an EAC of **USD 12,497,917** for a control account. The reviewer has the source data: budget at completion (BAC) **USD 11,900,000**, actual cost (AC) **USD 6,300,000**, earned value (EV) **USD 5,950,000**. The tool states no method.

Step 1 — reconstruct by the standard method. The cost performance index and the CPI-based EAC:

- `CPI = EV ÷ AC = 5,950,000 ÷ 6,300,000 = 0.9444`
- `EAC = AC + (BAC − EV) ÷ CPI = 6,300,000 + (11,900,000 − 5,950,000) ÷ 0.944444…`
- `= 6,300,000 + 5,950,000 ÷ 0.944444… = 6,300,000 + 6,300,000 = USD 12,600,000`

Rounding note. Using the four-decimal index instead of the unrounded one gives `5,950,000 ÷ 0.9444 = 6,300,296`, an EAC of `USD 12,600,296` — a rounding difference of USD 296. State which convention was used; do not let a rounding difference be mistaken for a method difference.

Step 2 — quantify the gap. `12,600,000 − 12,497,917 = USD 102,083`. The tool's figure is USD 102,083 lower than the cumulative-CPI method.

Step 3 — find the method. Solving for the index the tool must have used: `(BAC − EV) ÷ (EAC − AC) = 5,950,000 ÷ (12,497,917 − 6,300,000) = 5,950,000 ÷ 6,197,917 = 0.9600`. That is not the cumulative CPI of 0.9444; it is the three-period rolling CPI, which the reviewer confirms from the period data.

Result. The tool applied a **three-period rolling CPI of 0.9600** rather than the cumulative 0.9444. The difference is a method choice, not an error.

Interpretation. The reconstruction has converted an unexplained number into a professional judgement that can be made, recorded and defended. Which index basis is right depends on whether the variance driver is recent and closed or systemic and continuing — that judgement belongs to `AIG-04 — AI-assisted cost forecasting` and `BPG-09 — Estimate at completion`, and it is exactly the judgement an unreconstructed number silently makes on the professional's behalf. What the record must now carry is the method identified, the basis chosen, the reason, and the USD 102,083 the choice is worth. Note also what the reconstruction cost: three lines of arithmetic. Numbers go out unreconstructable because nobody asked, not because reconstruction was hard.

11. Checklist

- [ ] The project knows which of the three explanation depths (§2) each tool actually provides — and does not confuse driver attribution with reconstruction.
- [ ] A reconstruction record with the fields in §3 exists for every material AI-assisted output.
- [ ] Records are written at the time, not assembled on challenge, and retained to the records schedule.
- [ ] Every record names the data extract date, and Tier 1 work retains the extract.
- [ ] Prompts and configuration are captured for assistant and retrieval tools.
- [ ] Every difference between model output and reported figure is recorded with its reason.
- [ ] Sign-off names an individual.
- [ ] The evaluation set is segmented on project type, contract form, value band, region and stage, and segment results are recorded with the acceptance decision.
- [ ] The evaluation set is versioned and is never used to tune the model it tests.
- [ ] Drift re-runs have a named owner, a cadence and a trigger list; a score change is investigated before any re-tuning.
- [ ] Where reconstruction is unobtainable, the tool is used only as screen or corroboration and the record says so.

The test is not whether your organisation trusts the tool. It is whether, eleven months from now, a person who was not in the room can rebuild the number from what you wrote down.

Related

- `AIG-08 — Governing AI on a project — the control framework` — the register, tiers and change control this record sits inside
- `AIG-10 — Human in the loop: what AI may and may not decide` — why an unreconstructable output cannot carry a decision
- `AIG-04 — AI-assisted cost forecasting` — the forecasting methods this document holds to account
- `BPG-09 — Estimate at completion — choosing and defending a method` — the underlying method choice, without AI in the picture
- `CMP-08 — Data, digital and AI competencies in depth` — the competence a reconstruction presupposes

Sources and standards

- PCI Body of Knowledge, Domain 6 (Earned Value Management & Forecasting) and Domain 13 (AI for Project Controls & Project Management), `docs/bok/` — explained in our own words, not reproduced.

Where AI influences decisions about people — recruitment, allocation, performance — data-protection and employment law apply and vary by jurisdiction. This document describes the controls discipline; it is not legal advice, and it does not state any jurisdiction's requirements.

Status and version

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AIG-10 · AI IN PROJECT CONTROLS GUIDE

Human in the loop: what AI may and may not decide

The decision boundary, the tests that generate it, and the authority mechanics that make it hold

VERSION 1.0 · DRAFT

2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

Summary

"Human in the loop" is an authority statement, not a posture. This document distinguishes the three positions a human can actually hold, gives five tests that place any controls decision on the human side of the line, sets out the decision boundary those tests generate, and covers the mechanics that make it hold in practice — delegated authority thresholds, what a signature means, how rubber-stamping starts and is detected, and what changes when AI runs several steps in a chain.

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Contents

Human in the loop: what AI may and may not decide

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Who this is for. Project controls managers, cost and planning leads, change board members and project directors who hold delegated authority, and the practitioners whose names go on AI-assisted outputs.

1. The phrase does no work on its own

"There is a human in the loop" is offered as an assurance in almost every AI deployment discussion, and on its own it assures nothing. A human reading a screen is in the loop. A human clicking "approve" on forty items in six minutes is in the loop. A human who could not reproduce the number if asked is in the loop.

The phrase becomes a control when it answers three questions: **which decisions** does the human make rather than confirm; **at what threshold** does the decision move to someone more senior; and **what must the human have done** before their signature is worth anything. This document answers those three, in that order.

The discipline-specific boundaries are owned elsewhere in this series: `AIG-04 — AI-assisted cost forecasting`, `AIG-05 — AI in scheduling` and `AIG-06 — AI for risk identification and quantification`. What follows is the general apparatus that generates those lists, and the authority mechanics none of them covers.

2. Three positions, only one of which is what people mean

Position	What it means operationally	Where it is acceptable
In the loop	The system proposes; a human evaluates and decides; nothing proceeds without that decision	Any output meeting a test in §3
On the loop	The system acts; a human monitors, samples and can intervene	High-volume, low-consequence work with a measured error rate and an active monitoring duty (Tier 3, `AIG-08` §4)
Out of the loop	The system acts; no human reviews	Only where the action is trivially reversible with no external reliance or contractual effect — in controls, almost nothing

The distinction is not philosophical. In the loop costs review time per item and is the only position that gives you a decision-maker. On the loop costs monitoring effort and gives you a detection lag: errors happen, and you find them at the sampling frequency. Choosing "on the loop" accepts a known error rate — a legitimate professional choice, and only legitimate when the rate has been measured and the acceptance recorded.

Most deployments described as "human in the loop" are, on inspection, on the loop. That is often the right design. It becomes a governance failure when the framework claims the first and operates the second, so the monitoring duty that makes "on the loop" safe is never resourced.

3. Five tests that place a decision

Apply these to any controls decision an AI tool touches. **One positive answer puts the decision on the human side of the line.** They are deliberately blunt, because a boundary that requires fine judgement to apply will not survive a busy month.

1. Is it irreversible within the reporting cycle? A baseline changed, a contingency released, a notice served, a payment certified — none can be quietly undone next period. Reversibility, not size, is the first discriminator.

2. Does it have contractual or legal effect? Anything that asserts, concedes, waives or triggers a right. See `AIG-07 — AI in document control and correspondence` §3 for how this plays out in writing.

3. Will someone outside the project rely on it? A board, client, lender, auditor or certifier. External reliance converts an internal analysis into a representation, and representations are made by people.

4. Is the question fundamentally why, and contested? Attribution of cause — who caused the delay, whether escalation was foreseeable, whether rework is a defect or a variation — is the substance of most disputes. A model can assemble the evidence; it cannot hold a position.

5. Does it affect a person? Allocation, performance assessment, selection, capability judgement. This attracts data-protection and employment obligations that vary by jurisdiction, and a decision made about someone by a system nobody can explain is indefensible in every one of them.

One gating condition sits on top of the tests: **an output that cannot be reconstructed cannot carry any of these decisions**, whatever its apparent quality — see `AIG-09` §8.

4. The boundary

The tests generate the following. This is the function-level list; the discipline documents refine it.

Decision	AI may	A competent human must
Baseline change	Assemble the change pack, quantify time and cost impact, check the register for related items	Decide whether it is approved, on what scope and value, and sign the baseline revision (`BPG-04`)
Contingency release	Match the drawdown to the risk that materialised, compute remaining balance and burn rate	Decide the release, its amount and the authority level it requires (`BPG-10`, §5)
Extension of time entitlement	Assemble the chronology, extract notices and windows, run delay analyses on stated assumptions	Decide the entitlement position, the analysis method, and what is claimed or resisted (`BPG-12`)
A forecast submitted to a board or client	Produce candidate forecasts, decompose movement, flag inconsistency with the schedule	Choose the basis, state the assumption, own the number and sign (`AIG-04`)
Revenue recognition, provisioning, accrual judgement	Draft the working, run consistency checks, map facts to the policy	Make the judgement against the standard and the entity's policy, with the accounting owner
Acceptance of a subcontractor's programme or valuation	Check logic, compare against contract dates, reconcile quantities	Accept, reject or accept with reservation — acceptance is a commercial act
The contingency level adopted	Run the simulation, rank drivers, present the distribution	Choose the confidence level consistent with the organisation's appetite, and justify it (`AIG-06`)
What is disclosed, and how it is framed	Draft, check internal consistency, flag omissions	Decide what is said, what is caveated and what is not said
Decisions about people	Nothing that scores or ranks individuals without a specific, lawful, explainable basis	Decide, on evidence they can explain to the person affected
Changing a rule of credit or measurement basis	Model the effect of the change	Decide it, and disclose the discontinuity in the reported trend

Two things this table is not. It is not a list of tasks to keep AI away from — every row has a substantial AI contribution in the middle column, and refusing that contribution is its own failure. And it is not a maturity statement that relaxes as tools improve: the tests in §3 concern the nature of the decision, and a more capable model does not make a contested entitlement position less contested.

5. Thresholds: writing authority so it binds

A boundary without thresholds collapses into either paralysis or nothing. Delegated authority is the mechanism, and three rules make it work with AI in the picture.

Use the project's existing schedule. Do not invent an AI-specific authority ladder. AI-assisted decisions run through the same delegated authority as any other, expressed the same way — a value, a percentage of budget at completion (BAC), or a category. A parallel ladder guarantees divergence and gives an approver two answers to choose between.

Express the threshold on the decision, not the tool. "Contingency releases up to 0.25 % of BAC: controls manager" is a control. "AI-assisted releases: controls manager" is not, because it invites splitting a decision into an AI-assisted part and a human part to land in a lower band.

Name the aggregation rule. Thresholds are defeated by division. Related releases within a period, or against the same risk, aggregate for the threshold — and the rule is written down, because the split is rarely deliberate and always available. §10's worked example turns on this.

Add one AI-specific provision, and only one: **the approver must be able to state the basis of the proposal in their own words.** If they cannot, it is not ready for approval — a rule that is cheap to apply and catches the rubber-stamp before it becomes habitual.

6. What a signature actually asserts

A signature on an AI-assisted output asserts four things, and these are what it will be tested against.

1. **The inputs came from source**, not from the model's restatement of them.
2. **The method is known and appropriate**, and the reason for choosing it can be stated.
3. **The assumptions are stated** beside the answer, so a reader can challenge them.
4. **Differences between the model's output and the signed output are recorded**, with reasons.

"The model produced it" is not a defence, and neither is "the tool is approved" — the register governs which tools may be used and says nothing about whether this output is right. The verification record of `AIG-09` §3 is what the signature rests on.

The corollary is uncomfortable and worth stating plainly: **a reviewer who could not have produced the output by another route is not in a position to sign it.** Not every signatory must redo every calculation, but the capability must exist in the review, and where it does not the output goes to someone who has it. This is why the deskilling problem in `AIG-12 — The AI-literate controls professional` is a governance problem, not a training-department problem.

7. How rubber-stamping starts

It does not start with negligence. It starts with a run of correct outputs. The tool is accurate for several cycles, review finds nothing, and review time falls because nothing rewards it. The reviewer begins checking that the output looks reasonable rather than that it is right — a much faster operation, indistinguishable from the outside. Then an output is wrong in a way that looks reasonable, and it passes.

Three detections, none expensive.

Review time per output. If time booked to verification has fallen materially while volume has not, that is the signal. It is measurable if verification is a costed line (`AIG-08` §10), invisible if not.

Change rate. If no AI-assisted output has been materially changed in review for several cycles, either the tool is excellent or the review is not happening. Both are worth knowing; only one is being assumed.

Spot reconstruction. Once a quarter, pick a signed output at random and ask its signatory to reconstruct it (`AIG-09` §1). It is the only test that distinguishes the two cases above, and it takes under an hour.

8. When AI runs several steps

Chained and agentic systems — where the tool retrieves the data, computes, drafts and assembles without stopping — change the shape of the control, not the principle.

A single-step tool that errs produces one wrong output and one verification catches it. A chain that errs at step two carries the error forward, and every later step builds on it plausibly. The error is not merely propagated; it is polished. By the final output, the workings that would have exposed it are gone.

The response is to move verification **from per-output to per-workflow**. The professional assures the design of the chain — which steps are permitted, which data each may touch, where it must stop — and inserts checkpoints where a consequential intermediate result is inspected before the chain continues. Sensible checkpoints in controls: after extraction and before computation (is this the right cut?), after computation and before narrative (do the numbers reconcile?), and before anything leaves the function.

The audit trail must now record the chain, not just the answer. The decisions in §4 do not move: a chain may prepare a contingency release pack end to end, and the release is still decided by the person with the authority.

9. How this goes wrong

The loop with no decision in it. The workflow includes a human step, and that step is confirmation rather than decision. Nobody has ever declined. The control exists on the diagram.

Authority set on the tool rather than the decision. The threshold reads "AI-assisted analyses require manager approval". A decision is split into a model-produced analysis and a human-entered figure and lands below the line without anyone intending it.

Thresholds defeated by division. Four contingency releases of 90,000 against the same risk in one month, each within a lower authority. No aggregation rule, no visibility, no decision at the level the total warranted.

Signature without capability. The signatory cannot recompute and has no access to someone who can within the reporting timetable. The signature is a formality and everyone involved knows it.

Boundary rewritten by capability claims. A tool improves, and the argument follows that entitlement or recognition judgements can now be automated. The tests in §3 concern the nature of the decision: nothing about a better tool makes a contested cause uncontested.

Reasonableness substituted for correctness. Six good cycles, then review becomes a glance — invisible until an output is wrong and reasonable at the same time.

Chains verified only at the end. The final pack is checked carefully and every intermediate step trusted. An error introduced at extraction is confirmed by everything built on it.

Escalation with no route. The framework says contested items escalate; nothing says to whom, by when, or what happens if the reporting deadline arrives first. Under pressure the item is signed.

10. Worked example — a contingency release and the authority it needs

Illustrative figures. One project, one month. Currency USD. Thresholds expressed as a percentage of budget at completion.

Setup. Project BAC is **USD 48,000,000**. Contingency held is **USD 2,400,000** (5.0 % of BAC: `2,400,000 ÷ 48,000,000 = 0.05`). The delegated authority schedule for contingency release reads: controls manager up to **0.25 % of BAC**; project director up to **1.00 % of BAC**; sponsor above that. An AI-assisted analysis matches a materialised risk to a proposed release of **USD 180,000** and produces the supporting pack.

Step 1 — convert the thresholds to money.

- Controls manager `= 0.25 % × 48,000,000 = 0.0025 × 48,000,000 = USD 120,000`
- Project director `= 1.00 % × 48,000,000 = 0.01 × 48,000,000 = USD 480,000`

Step 2 — place the decision. `180,000 > 120,000` and `180,000 ≤ 480,000`, so the release is the **project director's decision**, not the controls manager's.

Step 3 — apply the aggregation rule. Two releases of **USD 95,000** and **USD 70,000** were made earlier in the same month against the same risk. Aggregated: `95,000 + 70,000 + 180,000 = USD 345,000`. This is still within the director's band (`345,000 ≤ 480,000`) but it is now `345,000 ÷ 2,400,000 = 14.4 %` of the total contingency drawn against a single risk in one month — a fact the director should be told and would not have been told by three separate approvals.

Step 4 — what AI contributed. The tool matched the drawdown to the risk, computed the remaining balance (`2,400,000 − 345,000 = USD 2,055,000`) and the burn rate, and assembled the evidence. It did not decide that the risk had materialised, that the amount was right, or that the remaining contingency is adequate for the risks that have not.

Result. A director-level decision, made on an aggregated view, with a remaining contingency of **USD 2,055,000** against a residual exposure the director must now judge.

Interpretation. The arithmetic is trivial; the control is not. Without the aggregation rule of §5, three defensible individual approvals produce an undisclosed concentration — each approver within authority, nobody seeing the total. Note also what the example refuses to do: treat the model's confidence in the match as a reason to lower the authority level. The threshold is a property of the decision's consequence, and the consequence of an over-drawn contingency does not change with how the recommendation was produced.

11. Checklist

- [ ] The framework states, per use, whether the human is in the loop or on the loop — and where it is on the loop, the error rate is measured and monitoring has a named owner.
- [ ] Every decision an AI tool touches has been run through the five tests in §3.
- [ ] The boundary in §4 has been adapted to this project's decisions and written into the plan.

- [ ] Thresholds are expressed on the decision rather than on AI involvement, and use the project's existing delegated authority schedule.
- [ ] An aggregation rule exists for related decisions within a period.
- [ ] Approvers can state the basis of a proposal in their own words.
- [ ] Signatories know the four assertions in §6, and outputs go to someone who can verify them when the first reviewer cannot.
- [ ] Review time and change rate are visible, and a spot reconstruction is done each quarter.
- [ ] Chained workflows have named checkpoints, and the audit trail records the chain.
- [ ] Escalation names a person and a deadline, and says what happens when the reporting date arrives first.

The line is not drawn to keep AI out of the work. It is drawn so that when a decision is questioned, there is a person who made it, on grounds they can state — which is the only thing that has ever made a project controls number worth anything.

Related

- `AIG-08 — Governing AI on a project — the control framework` — the register, tiers and records this boundary is enforced through
- `AIG-09 — Bias, explainability and auditability` — why an unreconstructable output cannot carry a decision at all
- `AIG-04 — AI-assisted cost forecasting` — the forecasting-specific boundary these tests generate
- `BPG-04 — Baselining and baseline change control` — the change authority this document assumes exists
- `BPG-10 — Contingency and management reserve` — the underlying discipline behind the worked example

Sources and standards

- PCI Body of Knowledge, Domain 12 (Risk Management for Project Controls) and Domain 13 (AI for Project Controls & Project Management), `docs/bok/` — explained in our own words, not reproduced.

Delegated authority schedules, accounting judgements and obligations relating to decisions about people are governed by an organisation's own policies and by law that varies between jurisdictions. This document describes the controls discipline and is not legal or accounting advice.

Status and version

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AIG-11 · AI IN PROJECT CONTROLS GUIDE

Evaluating AI tools — a buyer's due-diligence guide

The six questions that separate tools, the trial that tests them, and the cost the licence quote leaves out

VERSION 1.0 · DRAFT
2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

Summary

A due-diligence guide for buying AI capability into a controls function, written by capability class and not by product. It gives the six questions that actually separate one tool from another — data residency, training on your data, version management, export, audit trail and failure behaviour — with the acceptable and evasive answers to each; the trial that tests a claim on your own data; the total cost the licence quote omits; the contract terms worth negotiating; and how to run a pilot that is allowed to fail.

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Contents

Evaluating AI tools — a buyer's due-diligence guide

In one paragraph. A due-diligence guide for buying AI capability into a controls function, written by capability class and not by product. It gives the six questions that actually separate one tool from another — data residency, training on your data, version management, export, audit trail and failure behaviour — with the acceptable and evasive answers to each; the trial that tests a claim on your own data; the total cost the licence quote omits; the contract terms worth negotiating; and how to run a pilot that is allowed to fail.

Who this is for. Heads of project controls, PMO leads and commercial managers who specify, evaluate or approve AI tooling, and the practitioners asked to trial it.

1. What you are actually deciding

A tool evaluation looks like a product comparison and is not one. It is three decisions taken together, and confusing them is the commonest reason a purchase disappoints.

The first is **which capability class fits the task** — extraction, retrieval-grounded answering, tabular analysis, forecasting, anomaly detection, drafting. Settle it before any vendor is contacted, because a tool from the wrong class demonstrates beautifully and fails in production. `AIG-02 — What AI actually does in a controls function` owns that decision.

The second is **whether your data can support it**, which `AIG-03 — Data readiness` owns and which determines more of the outcome than the tool choice will.

The third — this document — is **which instance, on what terms**. It is a procurement decision, and it is governed by the same discipline the function applies to any other contract: the claim is a hypothesis, the terms are where the risk actually sits, and the total cost is not the quoted price.

This guide names no products. Capabilities and terms change on the vendor's release cycle, so any recommendation would be out of date before it was read. What does not change is the question set.

2. Specify before you shop

Write four things down before the first demonstration. They take an hour and they change the conversation from "what can it do" to "can it do this".

The task, in your own words. Not "AI for cost control" but "propose a cost code for each incoming invoice line, at a confidence threshold, with low-confidence lines routed to a human queue".

The maximum data classification the tool will handle, from the project's classification map (`AIG-08` §5). This eliminates candidates faster than any feature comparison.

The verification tier the outputs will sit at (`AIG-08` §4). A Tier 1 use has requirements — version identification, reconstructable method — that a Tier 3 use does not, and a tool that cannot meet them is unsuitable regardless of how well it performs.

The measure of success, as a number, with the baseline it is measured against. "Reduces coding effort" is not a measure. "Codes at least 80 % of lines at 97 % precision or better, against a baseline of three minutes per line" is.

3. The six questions

These are the questions that separate tools. Product feature lists converge; the answers to these do not. For each, the acceptable answer, the answer that should worry you, and what to ask next.

3.1 Where is our data processed and stored, and under whose law?

Acceptable: named regions for processing and storage, named subprocessors, a commitment that the region does not change without notice, and a statement of which jurisdiction's law governs access to the data.

Worrying: "in the cloud", "globally distributed", "we use industry-standard providers", or a region commitment for storage with silence about processing.

Ask next: does inference run in the same region as storage? Which subprocessors see the data, and what happens if one changes? What is the notification period for a region change?

3.2 Is our data used to train or improve your models?

Acceptable: a contractual statement that customer data is not used for training, with any exception named; a description of retention periods for prompts, inputs and outputs; and whether the answer differs between plan tiers, because it very often does.

Worrying: "your data is safe with us", an answer that covers the base model but not fine-tuning, telemetry or "product improvement", or an assurance available only in a marketing page.

Ask next: is the commitment in the contract or in a policy the vendor can change unilaterally? Does it cover human review of samples for quality purposes? What is retained, where, and for how long after deletion is requested?

3.3 How are versions managed, and what happens when the model changes?

Acceptable: a version identifier exposed in the interface or the audit log; advance notice of model changes; the ability to remain on a version for a defined period; and published notes on behavioural change, not only feature releases.

Worrying: no version identifier at all, or "we always give you the latest and best model". Continuous silent updating is a legitimate product decision and it is disqualifying for Tier 1 work, because a number produced last quarter cannot be reproduced.

Ask next: what identifier appears in the record of an output? What notice do we get, and can we test before the change applies? See `AIG-09` §4 for why "version not exposed" limits how a number can be defended.

3.4 What can we export, in what format, and how quickly?

Acceptable: export of your source data, the outputs, the configuration, the mappings and rules you built, and the audit log — in a documented, machine-readable format, on demand, without a professional services engagement.

Worrying: export of raw data only. The configuration is where your investment sits: the coding rules, the templates, the trained mappings, the tuned thresholds. A tool you can leave without your configuration is a tool you cannot leave.

Ask next: can we run an export now, during evaluation, and inspect it? What is excluded? What happens to our data at termination, and within what period?

3.5 What audit trail does the tool produce, and can we read it?

Acceptable: per-output records showing input reference, instruction or parameters, model or version, timestamp, user, and any human edit — exportable and retained for a period you set.

Worrying: a usage log rather than an output log. Knowing that a user ran 40 queries on Tuesday is telemetry; knowing which inputs produced which output under which version is an audit trail.

Ask next: show a real record for a real output. Can we retain it beyond the vendor's default? Does it survive an export?

3.6 How does it fail?

Acceptable: it declines when it cannot ground an answer, returns a stated confidence or a "not found", surfaces when a source was unavailable, and errors visibly on a malformed or partial input.

Worrying: it always returns an answer. A tool that never says "I do not know" has not eliminated uncertainty; it has stopped reporting it, and it has transferred the entire detection burden to your reviewer.

Ask next: show it failing. Give it a question its sources cannot answer, an input outside its range, a document it has not indexed. What the tool does in those three cases tells you more about your review workload than any accuracy figure.

4. Test it on your own data

A demonstration on the vendor's data is not evidence. Build a small evaluation set — inputs from your own work whose correct answers have been established by your own professionals, prepared to the project's confidentiality rules (`AIG-08` §5) and kept under version control. Fifty to a few hundred cases is usually enough to separate candidates.

Score every candidate on the same set, and record the results with the acceptance decision. Three disciplines make the trial worth running.

Segment the results. An aggregate score hides the failure that matters — good on the majority case, poor on the minority case that is often the project in front of you. See `AIG-09` §6.

Include the hard cases deliberately. Not the clean examples: the ambiguous invoice, the badly scanned drawing, the contract with a bespoke amendment, the project type you do least often.

Keep the set. It becomes the re-validation instrument for the life of the tool (`AIG-08` §6), and it is never used to tune the tool it tests.

5. Total cost, honestly netted

The licence quote is one line of five. A value case built on it will be approved and will not be true.

Cost	How to get it
Licence or metered usage	The quote — the only figure most business cases contain
Integration	Engineering effort to connect the systems the data lives in, plus ongoing maintenance
Data preparation	Cleaning, coding, structuring — usually the largest single item in year one (`AIG-03`)
Training	Everyone who will use it, plus refresher time
Governance and verification	The reviewer hours the framework requires, priced in `AIG-08` §10

The last line is the one vendors' business cases omit and the one that persists for the life of the tool. Netting it does not usually destroy a value case; it changes what is claimed. See `AIG-01 — AI in project controls — the executive guide` for the director-level version.

6. Terms worth negotiating

Six clauses, in the order they usually matter. This is contract discipline turned on the function's own purchases.

1. **Data residency and processing location**, with notice before change.
2. **No training on customer data**, stated in the contract rather than a policy, with any exception named.
3. **Intellectual property in outputs and configuration** — who owns the outputs, and who owns the mappings, rules and prompts built on your data.
4. **Export and exit** — format, scope (including configuration), timescale, cost, and deletion on termination.
5. **Notification of material model change**, with a defined notice period and a testing window.
6. **Audit rights or third-party assurance**, proportionate to the data classification the tool handles.

Lock-in through un-exportable configuration is priced at signature or discovered at exit. There is no third option.

7. What not to evaluate on

Demonstration polish. A rehearsed performance on curated data tells you the tool can do the task once, under favourable conditions, with the vendor driving.

The roadmap. Buy what exists. A committed future feature is worth its contractual remedy, which is usually nothing.

The reference customer list. Ask a reference what went wrong, not whether they are happy; only the first question is informative.

Benchmark scores. Published accuracy figures were measured on someone else's data and task definition. Your evaluation set is the only benchmark that speaks to your decision.

Whether it uses the newest model. The model is the least durable attribute of the product, and the one most likely to change without you.

8. Piloting, and letting a pilot fail

A pilot enters the permitted-use register as Provisional, with a success measure defined in advance (§2), an end date, a named owner and a defined data scope (`AIG-08` §3).

Two rules make a pilot worth running.

The measure is set before the pilot starts and is not revised during it. A measure adjusted mid-pilot to accommodate the result has stopped being a measure.

A pilot that misses its measure is closed, not extended. This is the rule that costs organisations the most to keep, because by the time a pilot ends someone senior has been associated with it publicly. The function that can close a failed pilot cleanly, record what it learned and say so is the function whose next business case will be believed.

Record the closure and the reason. A closed pilot with a documented reason is an asset: it stops the same idea returning in eighteen months with the same flaw and a different vendor.

9. How this goes wrong

The class was wrong and the evaluation could not tell. A general assistant is evaluated for a task needing a grounded retrieval tool. It demonstrates well — general assistants demonstrate well at everything — and hallucinates in production.

Residency answered for storage only. The contract commits to storage in a named region; inference runs elsewhere. Nobody asked the second question, and commercially sensitive material has crossed a border nobody agreed to.

Training exclusion in the wrong document. The assurance lives in a policy page the vendor can revise, not in the agreement. It was true when it was read.

A silent upgrade in the middle of a reporting cycle. Outputs shift, nobody was notified, and the figures either side of the change are not comparable. There was no version identifier to notice it by.

Export tested at exit. The data comes out; the configuration does not. Two years of coding rules and tuned thresholds stay with the vendor, and the migration cost exceeds the remaining licence cost.

The tool that never declines. It answers every question, so the reviewer stops expecting it to be unable to, and an ungroundable answer passes because it looked like every other answer.

The business case that counted the licence. The purchase is approved on the quote. Data preparation and verification effort arrive unbudgeted, verification is treated as optional, and the governance the value case implicitly assumed does not happen.

The pilot that could not fail. The measure was never numeric, the end date passed, and the tool is now in production because closing it would have been awkward.

10. Worked example — the total cost the quote did not show

Illustrative figures. Currency USD; a loaded internal labour rate is assumed; one tool for one controls function; three-year horizon; integration amortised straight-line over three years; rounding to the nearest whole currency unit.

Setup. A vendor quotes **USD 480 per seat per year** for **40 seats**. Internal estimates: integration **USD 35,000** one-off; data preparation **240 hours** in year one and **60 hours** in each later year; training **40 people × 4 hours**; verification effort **102 hours a year** (from `AIG-08` §10). Loaded labour rate **USD 85 per hour**.

Formulae. `licence = seats × price per seat`; `labour cost = hours × rate`; `amortised integration = one-off ÷ 3`.

Substitution — year one.

- Licence `= 40 × 480 = USD 19,200`
- Integration `= 35,000 ÷ 3 = USD 11,667` (11,666.67 rounded)
- Data preparation `= 240 × 85 = USD 20,400`
- Training `= 40 × 4 × 85 = USD 13,600`
- Verification `= 102 × 85 = USD 8,670`
- **Year one total** `= 19,200 + 11,667 + 20,400 + 13,600 + 8,670 = USD 73,537`

Substitution — years two and three (each).

- `= 19,200 + 11,667 + (60 × 85 = 5,100) + 8,670 = USD 44,637`

Three-year total. Computed from the unamortised components: `(19,200 × 3) + 35,000 + (20,400 + 5,100 + 5,100) + 13,600 + (8,670 × 3) = 57,600 + 35,000 + 30,600 + 13,600 + 26,010 = USD 162,810`. Rounding note: summing the three annual figures gives `73,537 + 44,637 + 44,637 = USD 162,811`, one dollar higher, because the amortised integration line rounds up each year. Quote one basis and say which.

Result. The quoted licence of **USD 19,200** is **26.1 %** of year-one cost (`19,200 ÷ 73,537 = 0.261`) and **35.4 %** of the three-year cost (`57,600 ÷ 162,810 = 0.354`).

Interpretation. Nothing here argues against the purchase. It argues against approving it on the quote. Three consequences follow. The approval goes to whoever can approve USD 162,810, not USD 57,600. The benefit side must clear a bar nearly four times the licence in year one, which is a genuine test and usually a passable one. And the two internal lines — data preparation and verification — are the ones that do not appear unless someone puts them there, which is why they are also the two that are silently cut when the year gets difficult. Note what is not in the table: the cost of the tool being wrong. That is priced separately, and it is why the verification line exists rather than being an overhead to trim.

11. Evaluation scorecard

Score each candidate; an unanswered question is a scored answer.

- [ ] The capability class was decided before the vendor was contacted (`AIG-02`).
- [ ] The task, maximum data classification, verification tier and numeric success measure are written down.

- [ ] Processing and storage regions are named, with subprocessors and notice on change.
- [ ] The no-training commitment is in the contract, with exceptions named and retention periods stated.
- [ ] A version identifier appears in the output record, with advance notice of model change and a testing window.
- [ ] An export has been run during evaluation and inspected — data, outputs, **and** configuration.
- [ ] A real per-output audit record has been shown, and it survives export.
- [ ] The tool has been observed failing on an ungroundable question, an out-of-range input and an unindexed document.
- [ ] It has been scored on your own versioned evaluation set, segmented, including the hard cases.
- [ ] The cost case includes integration, data preparation, training and verification, not the licence alone.
- [ ] The six contract terms in §6 have been raised, and the answers recorded.
- [ ] The pilot has a numeric measure, an end date, a named owner and a stated data scope — and permission to fail.

The question that ends most evaluations honestly is the last one in §3.6: ask to see it fail. A vendor who can show you that is telling you what your reviewers will actually spend their time on, which is the only number in the whole exercise that you will still be living with in two years.

Related

- `AIG-02 — What AI actually does in a controls function` — the capability class decision that precedes any tool choice
- `AIG-03 — Data readiness: what AI needs before it is any use` — the constraint that determines the outcome more than the tool does
- `AIG-08 — Governing AI on a project — the control framework` — the register, data classes and tiers a purchased tool must fit into
- `AIG-09 — Bias, explainability and auditability` — why version exposure and audit records are evaluation criteria, not nice-to-haves
- `TPL-15 — Project controls health check` — the wider assurance instrument this evaluation feeds

Sources and standards

- PCI Body of Knowledge, Domain 7 (Contracts, Commercial Management, BoQ, Invoicing & Revenue) and Domain 13 (AI for Project Controls & Project Management), `docs/bok/` — explained in our own words, not reproduced.

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AIG-12 · AI IN PROJECT CONTROLS GUIDE

The AI-literate controls professional

Six capabilities, the five questions that interrogate an AI-produced number, and how the underlying craft is kept alive

VERSION 1.0 · DRAFT
2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

Summary

AI literacy in project controls is not prompt technique. It is six observable capabilities — choosing when not to use AI, judging whether data is fit and safe, directing an output precisely, interrogating an AI-produced number, governing and signing it, and keeping the underlying craft alive. This document defines each behaviourally, gives the five questions that take an AI-produced number apart in the right order, sets out what evidence of literacy an employer can actually observe, and treats deskilling as the governance problem it is.

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Contents

The AI-literate controls professional

In one paragraph. AI literacy in project controls is not prompt technique. It is six observable capabilities — choosing when not to use AI, judging whether data is fit and safe, directing an output precisely, interrogating an AI-produced number, governing and signing it, and keeping the underlying craft alive. This document defines each behaviourally, gives the five questions that take an AI-produced number apart in the right order, sets out what evidence of literacy an employer can actually observe, and treats deskilling as the governance problem it is.

Who this is for. Cost engineers, planners, quantity surveyors and controls analysts developing their practice; the managers who assess them; and employers writing a role specification that has to mean something.

1. What AI literacy is not

Three things are routinely mistaken for it, and all three are cheap to acquire and worth little.

Prompt technique. Knowing phrasings that produce better output is useful and shallow. It transfers poorly between tools, dates quickly, and says nothing about whether the practitioner can tell a good output from a plausible one.

Tool familiarity. Having used several products is experience, not competence. The person who has used six tools and verified nothing has practised the wrong thing six times.

A position on AI. Enthusiasm and scepticism are both available without any underlying capability. The professional question is never whether AI is good; it is whether this output, for this decision, can be relied on, and the answer is arrived at by work.

What literacy actually consists of is unglamorous: knowing which tasks a class of tool can do, being able to tell whether an output is right, and being able to say why you believe it. That capability is domain-first. A practitioner who cannot forecast without AI cannot verify a forecast produced with it — which is the whole hinge of this document.

2. Six capabilities

Stated as behaviour, because a competency written as knowledge cannot be assessed.

1. Choose — including choosing not to. Matches the task to a capability class (`AIG-02`) rather than reaching for the tool at hand, and declines AI where a deterministic rule is more transparent, where confidentiality cannot be assured, where the data will not support it, or where verification would cost more than doing the work. Observable behaviour: the practitioner can name a task in their own work that they deliberately do not use AI for, and give the reason.

2. Prepare — judge whether data is fit and safe. Assesses whether the data supports the question (`AIG-03`) and whether it may lawfully and contractually be entered into the tool proposed.

Observable behaviour: they check the data classification before the first paste, not after the first result.

3. Direct — specify an output precisely. Constructs an instruction containing the six elements in §4, including the instruction to decline rather than guess. Observable behaviour: their instructions state what to do when the answer is not in the source.

4. Interrogate — take an AI-produced number apart. Applies the five questions in §3 in order, and reaches a view about the number rather than about the tool. Observable behaviour: they can say what a figure would have to have been for the claim attached to it to be true — the discipline the worked example in §10 demonstrates.

5. Govern — verify, record and sign. Applies the verification tier, keeps the reconstruction record (`AIG-09` §3), records what changed in review and why, and knows the four things a signature asserts (`AIG-10` §6). Observable behaviour: an output they signed six months ago can be reconstructed from what they wrote down.

6. Maintain — keep the craft alive. Works problems by hand often enough to retain the fluency verification depends on, and can explain a method to a colleague without a tool in front of them. Observable behaviour: they can produce the calculation on a whiteboard.

Capabilities 4 and 6 are the ones that distinguish practitioners. The first three are learnable in weeks. The last two are built over years and lost quietly.

3. Interrogating an AI-produced number

Five questions, in this order. The order matters: each is cheaper than the one after it, and a failure at any point makes the rest unnecessary.

1. What are the inputs, and are they from source? Take the inputs from the controlled system, not from the model's restatement of them. Confirm the data cut. Most wrong answers are wrong here, and this check takes minutes.

2. Is it dimensionally and directionally sensible? Right order of magnitude, right sign, right units, right period. A forecast that moved the wrong way, a percentage over an unstated base, a figure that is a factor of ten out — these are visible before any method question and are missed because nobody looks.

3. What method produced it, and does the method match the situation? Ask which method, not whether the answer is right. A defensible method producing an uncomfortable answer is a finding; an undisclosed method producing a comfortable answer is a risk. Where the method is not stated, reconstruct it — `AIG-09` §10 shows this being done in three lines of arithmetic.

4. What must be true for this to hold? Name the assumptions the answer depends on, and test the load each one carries. The question that gets to it fastest: what would have to be different for this number to be materially wrong?

5. Does it reconcile with what I already know? Against the prior period, against the schedule, against commitments, against the physical work. A number that is internally coherent and inconsistent with the project is still wrong.

Only after all five is there a professional judgement to make — and that judgement is about the project, not about the model.

4. Directing an output

A precise instruction contains six elements. This is not prompt craft; it is specification, and the same elements would be required of a graduate given the same task.

Role and standard — the discipline perspective and the standard of care expected. **Task** — the one thing to produce. **Source** — the data or documents to work from, and the instruction to use nothing else. **Constraints** — length, format, tone, and what must not be included. **Output shape** — the table, fields or structure required, so the result is checkable rather than merely readable. **Refusal instruction** — what to do when the answer is not in the source: say "not found", do not infer, do not estimate.

The sixth element is the one practitioners omit and the one that most reduces risk. A model asked a question it cannot answer from the source will produce a plausible answer unless told not to. Adding "if it is not in the attached documents, say 'not found'" converts a silent fabrication into a visible gap.

5. What evidence of literacy looks like

For an employer writing a specification, or a manager assessing one, these are observable — unlike "familiar with AI tools", which is not.

Capability	Evidence a manager can actually see
Choose	Can name a task they decline to use AI for, with the reason
Prepare	Checks data classification before use; can say what preparation an analysis needed
Direct	Instructions include a source constraint and a refusal instruction
Interrogate	Has rejected or materially changed an AI output, and can say what the check was
Govern	A signed output from six months ago can be reconstructed from their record
Maintain	Can produce the underlying calculation without a tool

The single most informative question in an interview or an appraisal is: **"tell me about an AI output you rejected, and how you knew."** A practitioner who has never rejected one has either not been using AI or has not been checking it, and the follow-up question distinguishes those cases in under a minute.

6. Building it

A practice sequence that produces artefacts rather than attendance. Each item is small; the sequence is what does the work.

Recompute before you accept, for one month. Every AI-assisted number that crosses your desk, taken apart by §3 before it is used. This is slow and it is where the instinct comes from.

Keep an error log. Every AI output you found wrong, what the error was, and which of the five questions caught it. After twenty entries you will know which failure modes your tools actually have, which no general guidance can tell you.

Work one method by hand each month, unaided. A forecast, a variance bridge, a network pass, a reconciliation, built from source with no tool. This is deliberate practice, and its purpose is not the answer.

Write one specification a week. Take a task you did conversationally and write it out with the six elements of §4. Compare the outputs.

Verify something outside your specialism, with the specialist. A planner checking a cost forecast with a cost engineer learns the questions faster than either would alone.

Teach one method to someone junior, without a tool. Teaching is a severe test of understanding, and it is also how the next generation of verifiers is produced — see §7.

7. The deskilling problem

The governance model of this whole series rests on verification, and verification presupposes a verifier who can still do the work: recompute the forecast, spot the wrong assumption, recognise an unrealistic duration. Those instincts have always been built by doing the work — historically by junior practitioners doing precisely the tasks that AI now does first.

Today's verifiers trained before AI. The honest position is that where tomorrow's come from is an open question with no settled answer, and that a function which waits for the evidence to appear in its own error rates has waited too long. This is not an argument against using AI. It is an argument for producing deliberately what the daily workflow used to produce as a by-product.

Three counter-measures are within any function's reach, and all three are cheap relative to one undetected forecast error.

Rotation through first-principles work. Defined periods in which developing practitioners build an estimate, a schedule or a reconciliation from source, with the tools off, so judgement forms on the task rather than on reviewing a draft of it.

Verification taught as a skill in its own right. How to recompute, how to ground an extraction, how to challenge a causal claim — taught and assessed rather than assumed. A reviewer who only confirms is already deskilled.

Review capability tracked as a capacity constraint. If only two people in a function can genuinely verify a Tier 1 output, that is a resourcing fact with the same status as any other single point of failure, and it belongs on the risk register rather than in a training plan.

8. Where this sits in the Institute's scheme

The Institute treats AI literacy as part of the discipline, not as an adjacent subject. Two of the fourteen seeded competencies for the **PCI AI Project Controls Leader (PCL-AI)** credential are **Responsible AI** and **Human validation**, alongside AI-enabled project controls, predictive analytics and automation — they sit beside cost management and earned value rather than after them. A separate, open-entry offering, the **AI in Project Controls — Specialist Certificate (AIPC)**, is drawn from Domain 13 of the PCL-AI Body of Knowledge and is assessed by scenario items plus an applied AI task.

Currency is treated as a live obligation rather than a one-off. Recertification runs on a three-year cycle with a mandatory AI-currency component, and the continuing professional development category that satisfies it is named "AI currency" in the Institute's own records [CONFIRM: binding CPD hours per three-year cycle — the student portal shows a target of 30 hours, and the binding requirement will be published with the recertification rules]. The reason for treating currency as mandatory is in §7 rather than in any regulation: capability that is not exercised does not persist, and a credential that certified verification once is worth less each year that nobody checks whether the holder still can.

Detail on how the competencies are levelled and evidenced belongs to `CMP-08 — Data, digital and AI competencies in depth`; the recertification mechanics belong to `CER-08 — Recertification, CPD and the AI-currency requirement`.

9. How this goes wrong

Fluency mistaken for competence. The practitioner is quick, confident and productive with the tools, and has never taken an output apart. The gap is invisible until a plausible output is wrong.

Verification taught as a checklist. People learn to tick "source-checked" without knowing what source-checking a forecast involves. The form is completed and the check is not performed.

Interrogation stopped at question three. The method is confirmed, the assumptions are never named, and the number is signed with a load-bearing assumption nobody has stated.

The specialist who cannot be checked. One person owns the AI-assisted workflow, and nobody else in the function can verify it. This is presented as expertise and is a single point of failure.

Training on the tool instead of the discipline. The programme covers the product's features and is obsolete at the next release. Nothing was taught that survives a change of vendor.

Practice replaced by attendance. Hours are logged, artefacts are not produced, and no calculation was worked by hand. Currency is recorded and not held.

Rejection treated as obstruction. A practitioner who declines outputs is seen as slowing delivery. Within two cycles they stop, and the function loses the only signal it had that review was real.

10. Worked example — interrogating a causal claim

Illustrative figures. One control account, one period. Currency USD; figures cumulative to the data date; percentages stated against the variance as base.

Setup. An AI-drafted variance narrative states: "The unfavourable cost variance of USD 320,000 is approximately 80 % driven by rate escalation on structural steel." The reviewer applies question 5 of §3 — does it reconcile with what I already know — and has two facts from source: structural steel cost to date is **USD 1,600,000**, and the escalation applied per the procurement record is **4.5 %**.

Step 1 — compute the escalation effect. `escalation effect = steel cost to date × escalation rate = 1,600,000 × 0.045 = USD 72,000`

Step 2 — express it against the variance. `share = 72,000 ÷ 320,000 = 0.225 = 22.5 %`

Step 3 — test the claim from the other direction. For escalation to be 80 % of the variance: `0.80 × 320,000 = USD 256,000`, which would require an escalation rate of `256,000 ÷ 1,600,000 = 0.16 = 16 %` — three and a half times the rate in the procurement record.

Result. The claim is wrong. Escalation accounts for **22.5 %** of the variance, not 80 %. The remaining `320,000 − 72,000 = USD 248,000`, or **77.5 %** of the variance, is unattributed.

Interpretation. Note what the check cost: two multiplications and a division, using two figures the reviewer already had. Note also what it produced. It did not merely correct a narrative — it revealed that USD 248,000 of variance has no explanation, which is now the most important open item on the account and which the drafted narrative would have closed off. This is the difference between reviewing text and interrogating a number: the first fixes a sentence, the second finds the work. The assumptions this answer depends on are worth stating with it — that steel cost to date is complete at the data date, and that the 4.5 % in the procurement record is the rate actually applied rather than the rate agreed.

11. Checklist

Use this on yourself, or on a role specification.

- [ ] I can name a task in my own work that I deliberately do not use AI for, and say why.
- [ ] I check the data classification before the first paste, not after the first result.
- [ ] My instructions name the source and say what to do when the answer is not in it.
- [ ] I take inputs from the controlled system, never from the model's restatement of them.
- [ ] I apply the five questions of §3 in order, and I get past question three.
- [ ] I can name the assumptions any number I signed depends on.
- [ ] I have rejected or materially changed an AI output in the last quarter, and I can say what the check was.
- [ ] An output I signed six months ago could be reconstructed from what I wrote down.
- [ ] I work one method by hand each month, unaided.
- [ ] I keep a log of AI errors I have caught, and which question caught them.
- [ ] My function has more than one person who can verify a Tier 1 output.

The capability that matters is not the ability to get a good answer out of a tool. It is the ability to tell, quickly and for reasons you can state, that the good-looking answer in front of you is wrong — and that capability is built by doing the work, which is now something a profession has to arrange on purpose.

Related

- `AIG-09 — Bias, explainability and auditability` — the record that capability 5 produces, and the reconstruction discipline behind question 3
- `AIG-10 — Human in the loop: what AI may and may not decide` — what a signature asserts, and why review capability is a governance constraint
- `AIG-02 — What AI actually does in a controls function` — the capability classes capability 1 chooses between
- `CMP-08 — Data, digital and AI competencies in depth` — how these competencies are levelled and evidenced

- `CER-08 — Recertification, CPD and the AI-currency requirement` — the currency obligation referred to in §8

Sources and standards

- PCI Body of Knowledge, Domain 13 (AI for Project Controls & Project Management), `docs/bok/` — explained in our own words, not reproduced.
- The Institute's candidate AI-use policy (`docs/downloads/`) — the position on AI in preparation, in the examination and in professional practice.

Competency and credential facts in §8 are taken from the Institute's own seeded competency sets and published scheme documents. The continuing professional development requirement is not yet fixed and is marked as a placeholder rather than stated.

Status and version

Founding-stage document · Version 1.0 · draft · Reviewed under PCI governance. PCI makes no claims of accreditation or recognition beyond what is true today.